

Chapter 2

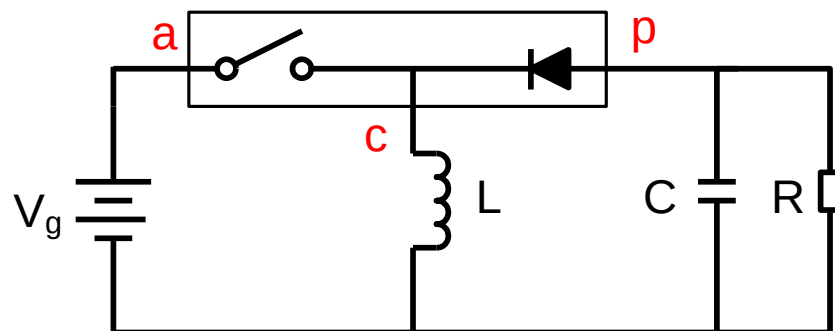
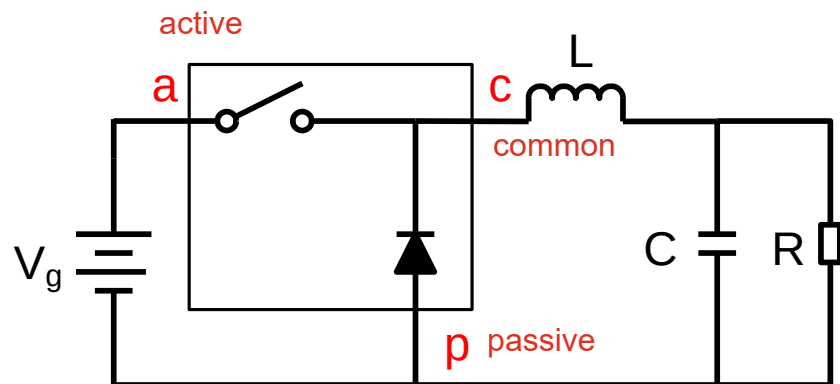
PWM Switch Model

Chapter 2.1

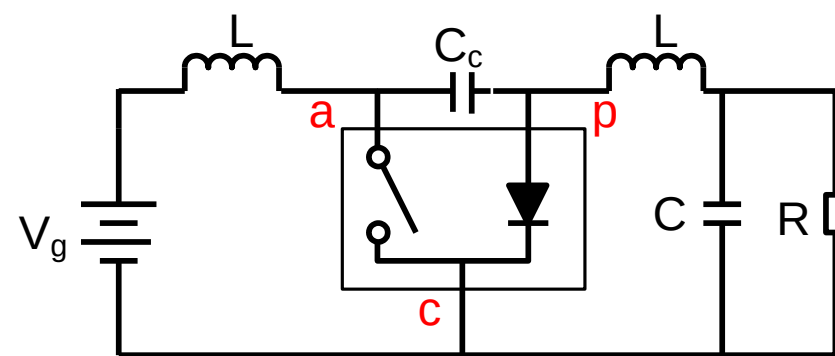
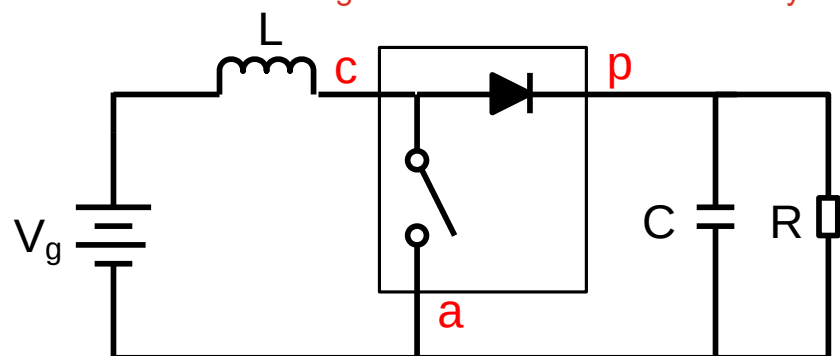
CONTINUOUS CONDUCTION MODE [1]

[1] Vatche Vorperian, "Simplified analysis of PWM converters using model of PWM Switch Part 1: Continuous Conduction Mode", IEEE Transactions on Aerospace and Electronic System, Vol.26, No.3 May 1990, pp. 490-496.

Basic Converter Cells

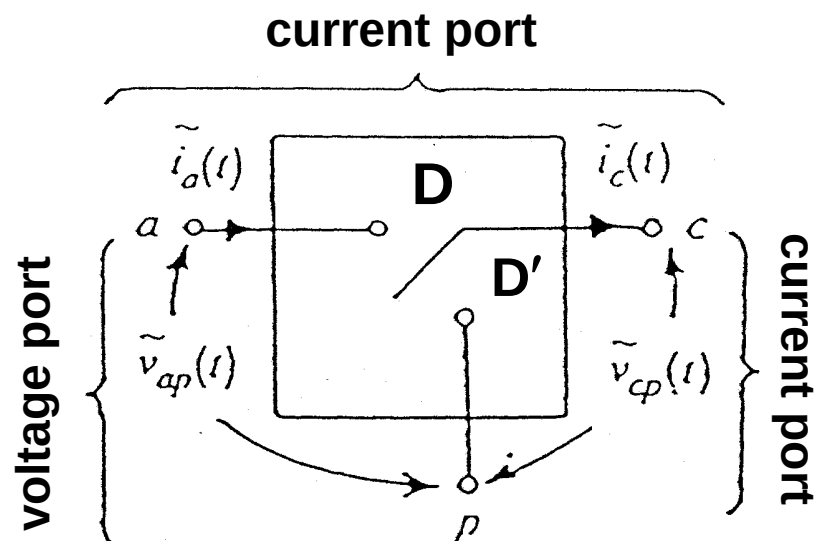


treat switch pair as a basic components
then the rest is passive components, small
signal model can be derived easily



1. a-p is connected to a voltage source, so c terminal can generate a DC+AC waveform
2. common terminal is always connected to a equivalent inductor path

Laws of interconnect



The PWM Switch

a = active terminal

p = passive terminal

c = common terminal

$\tilde{i}_a(t)$ = instantaneous active terminal current

$\tilde{i}_c(t)$ = instantaneous common terminal current

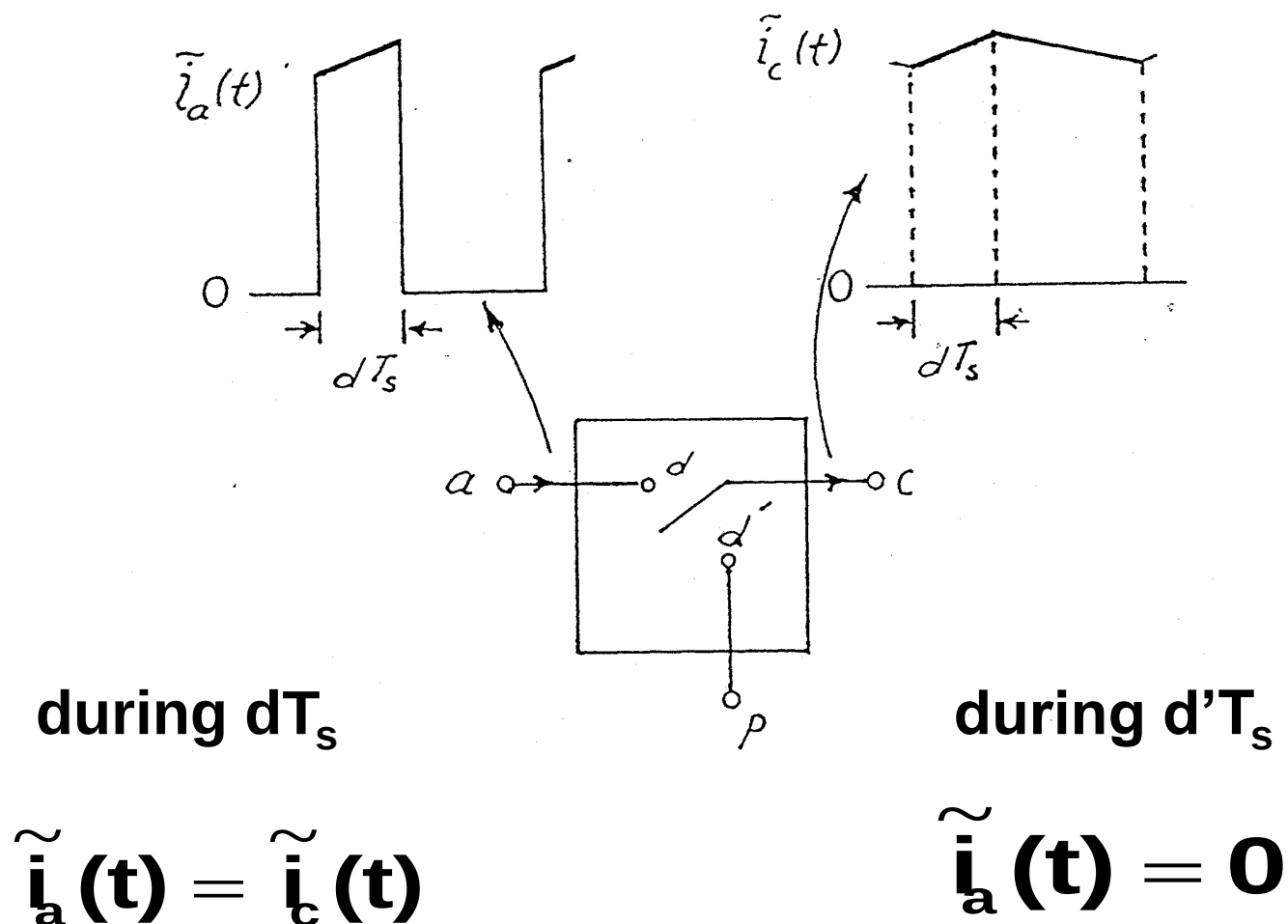
$\tilde{v}_{ap}(t)$ = instantaneous voltage across port *ap*

$\tilde{v}_{cp}(t)$ = instantaneous voltage across port *cp*

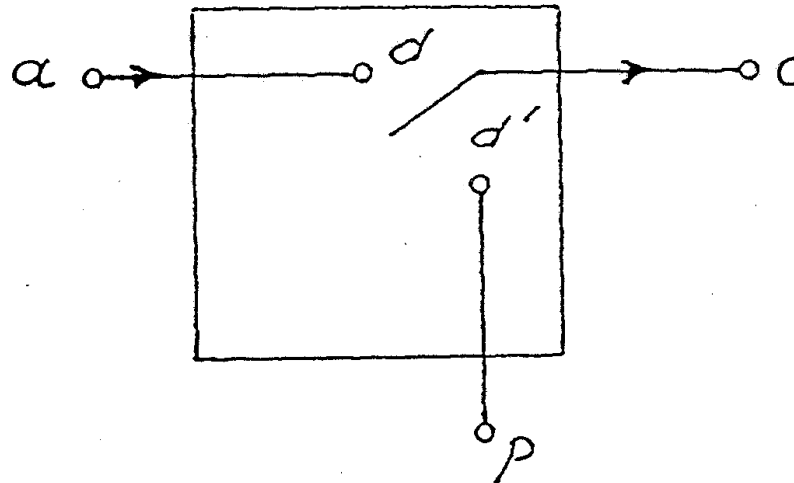
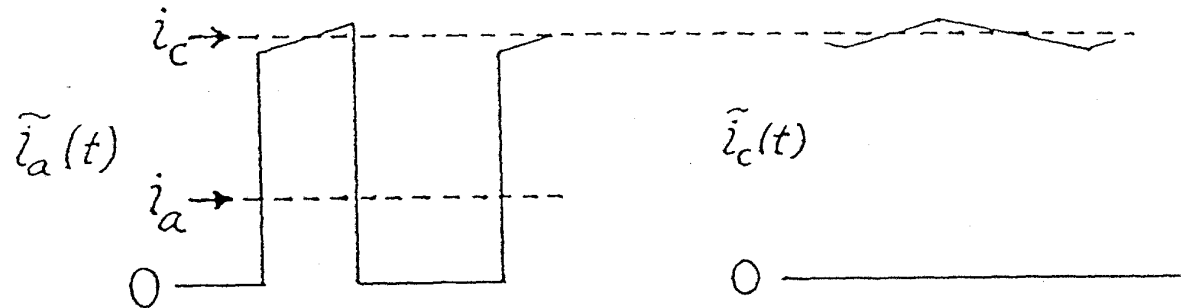
D = on-time duty cycle

D' = 1 – D

The Invariant Properties of the PWM Switch

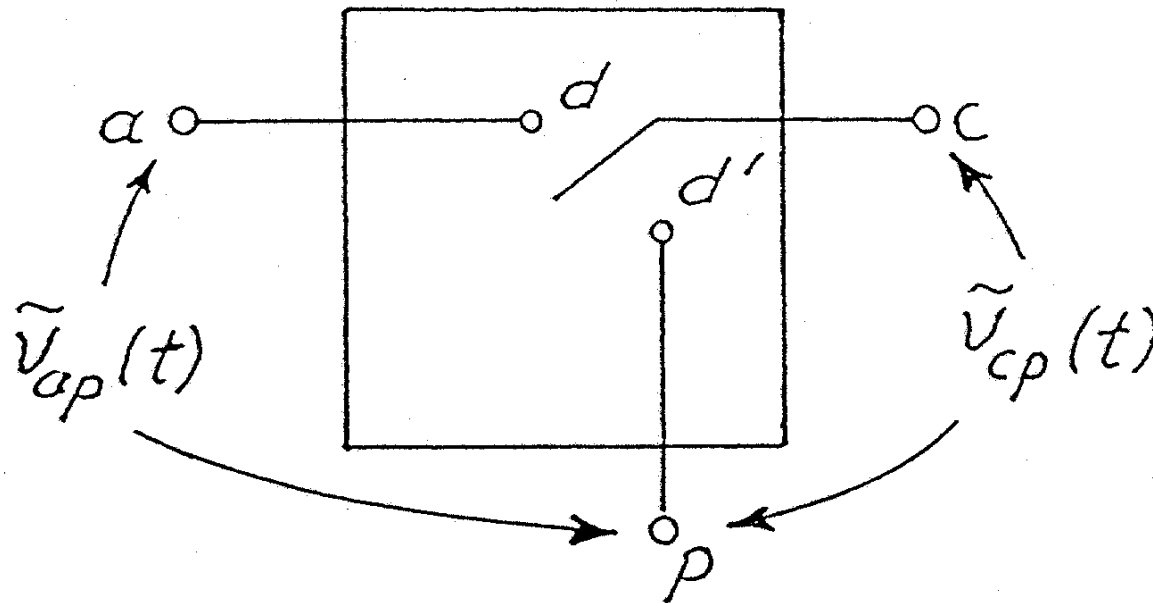


The Invariant Property in the average Sense



$$\mathbf{i}_a = \mathbf{d} \cdot \mathbf{i}_c$$

The Invariant Properties of the PWM Switch



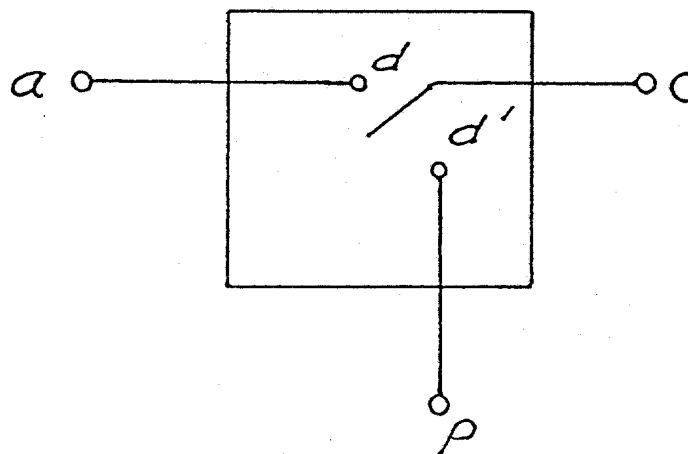
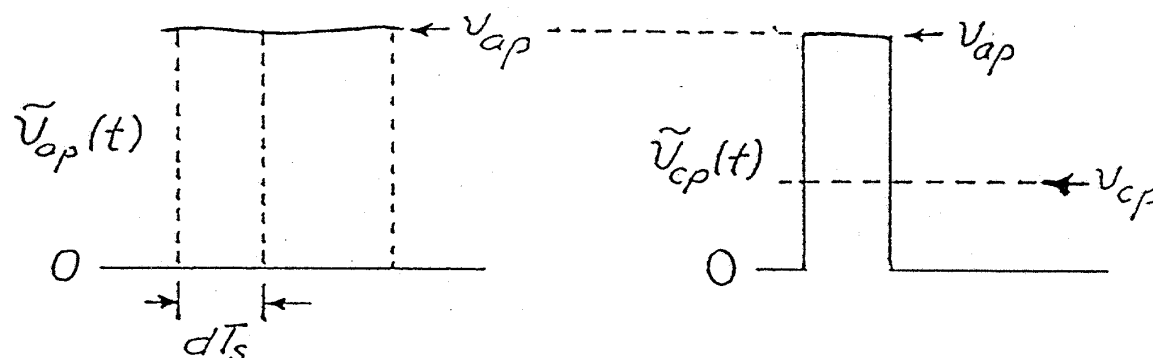
during dT_s

$$\tilde{\mathbf{v}}_{cp}(\mathbf{t}) = \tilde{\mathbf{v}}_{ap}(\mathbf{t})$$

during $d'T_s$

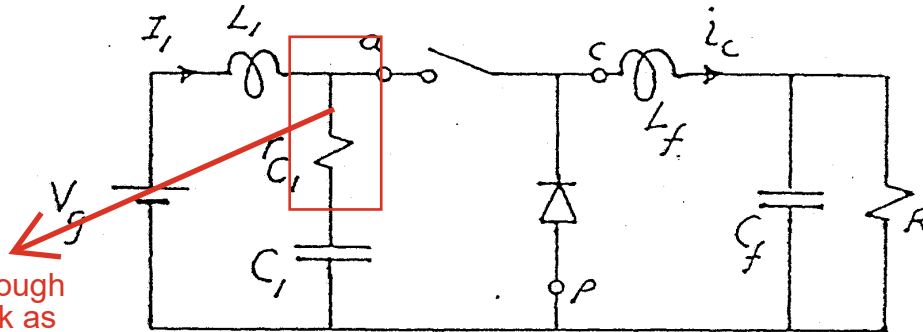
$$\tilde{\mathbf{v}}_{cp}(\mathbf{t}) = \mathbf{0}$$

Case 1: $\tilde{v}_{ap}(t)$ is continuous



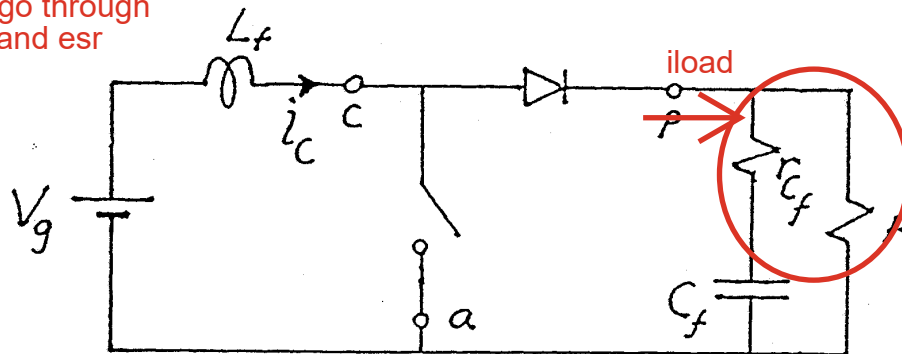
$$\mathbf{v}_{cp} = \mathbf{d} \cdot \mathbf{v}_{ap}$$

Case 2 : $\tilde{v}_{ap}(t)$ is discontinuous



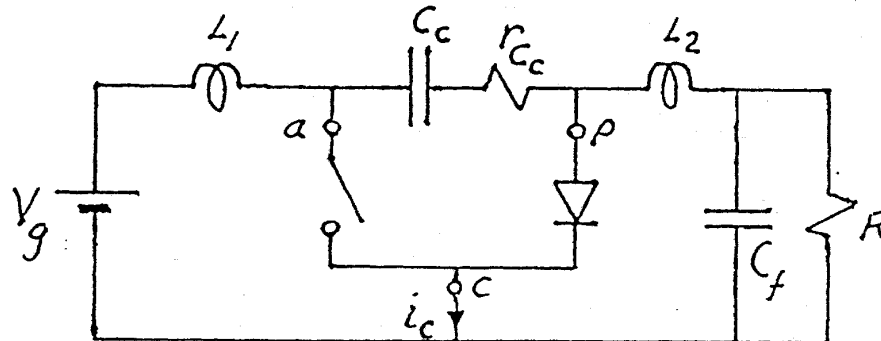
ac current go through
switching network as
small signal model go through
the input capacitor and esr

$$r_e = r_{C_1}$$



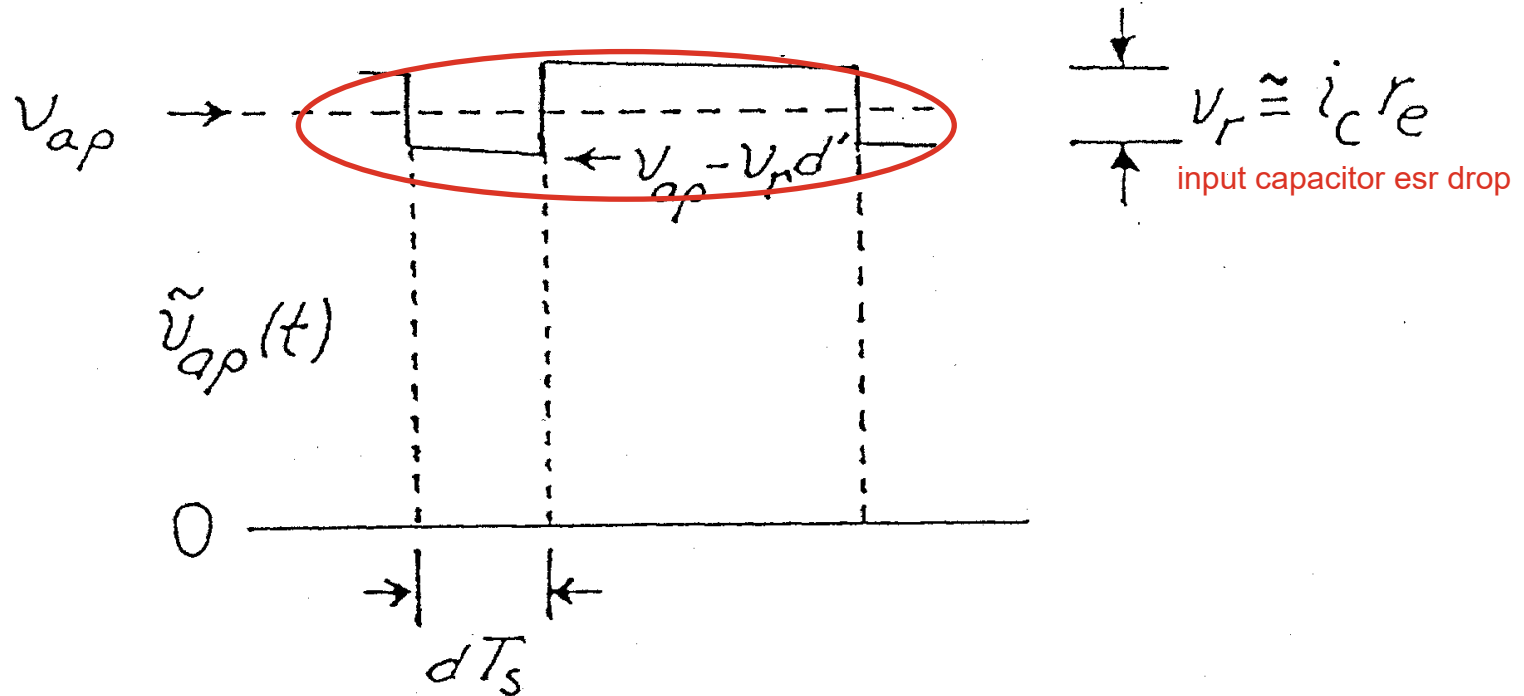
iload will have ripple, but it go through both
cap esr and load resistor, so $r_e = r_{esr} // R$

$$r_e = r_{C_f} || R$$



$$r_e = r_{C_c}$$

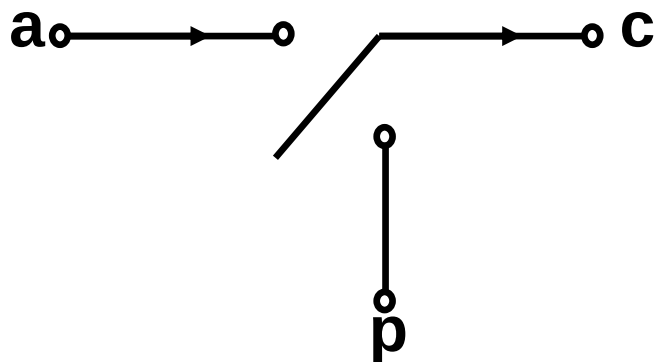
Case 2: $\tilde{\mathbf{v}}_{ap}(t)$ is discontinuous



$$\mathbf{v}_{cp} = \mathbf{d} \cdot (\mathbf{v}_{ap} - \mathbf{v}_r \cdot \mathbf{d}')$$

Derivation of DC and Small Signal Models for PWM Switch

Three-terminal Switch



(Large-Signal)

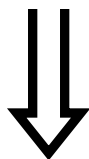
Average model:

$$i_a = d \cdot i_c$$

$$v_{cp} = d \cdot (v_{ap} - i_c r_e d')$$

Perturbation

$$\begin{aligned}
 \mathbf{i}_a &= \mathbf{I}_a + \hat{\mathbf{i}}_a & \mathbf{i}_c &= \mathbf{I}_c + \hat{\mathbf{i}}_c & \mathbf{d} &= \mathbf{D} + \hat{\mathbf{d}} \quad (\mathbf{d}' = \mathbf{D}' - \hat{\mathbf{d}}) \\
 \mathbf{v}_{cp} &= \mathbf{V}_{cp} + \hat{\mathbf{v}}_{cp} & \mathbf{v}_{ap} &= \mathbf{V}_{ap} + \hat{\mathbf{v}}_{ap} & \hat{\mathbf{i}}, \hat{\mathbf{v}} &\ll \mathbf{I}, \mathbf{V}
 \end{aligned}$$



$$\begin{aligned}
 \underline{\mathbf{I}_a} + \underline{\hat{\mathbf{i}}_a} &= (\mathbf{D} + \hat{\mathbf{d}}) \cdot (\mathbf{I}_c + \hat{\mathbf{i}}_c) \\
 &= \underline{\mathbf{D} \cdot \mathbf{I}_c} + \underline{\mathbf{D} \cdot \hat{\mathbf{i}}_c} + \underline{\mathbf{I}_c \cdot \hat{\mathbf{d}}} + \underline{\hat{\mathbf{d}} \cdot \hat{\mathbf{i}}_c}
 \end{aligned}$$

$$\begin{aligned}
 \underline{\mathbf{V}_{cp}} + \underline{\hat{\mathbf{v}}_{cp}} &= (\mathbf{D} + \hat{\mathbf{d}}) \cdot [(\mathbf{V}_{ap} + \hat{\mathbf{v}}_{ap}) - (\mathbf{I}_c + \hat{\mathbf{i}}_c) \cdot \mathbf{r}_e \cdot (\mathbf{D}' - \hat{\mathbf{d}})] \\
 &= \underline{\mathbf{D} \cdot \mathbf{V}_{ap}} - \underline{\mathbf{D} \cdot \mathbf{I}_c \cdot \mathbf{r}_e \cdot \mathbf{D}'} + \underline{\mathbf{D} \cdot \hat{\mathbf{v}}_{ap}} + \underline{(\mathbf{V}_{ap} + \mathbf{I}_c \mathbf{D} \mathbf{r}_e - \mathbf{I}_c \mathbf{D}' \mathbf{r}_e) \cdot \hat{\mathbf{d}}} - \underline{\mathbf{D} \mathbf{D}' \mathbf{r}_e \cdot \hat{\mathbf{i}}_c} + \dots
 \end{aligned}$$

— DC Component

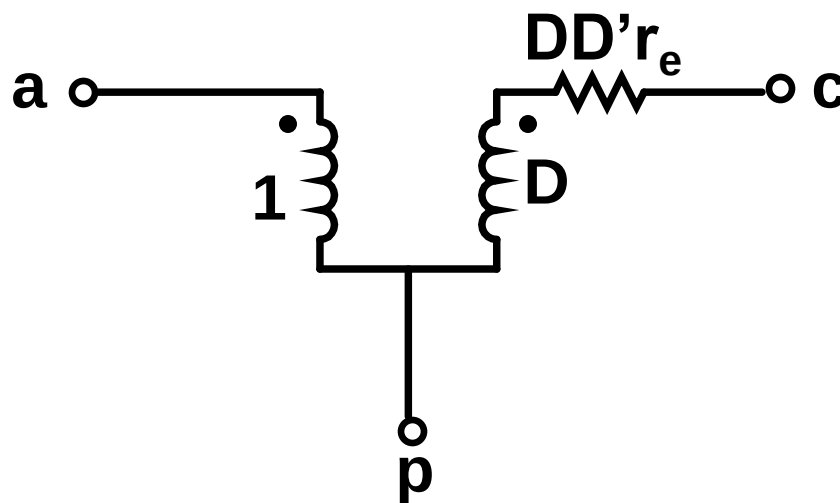
— Small-signal
Component

Higher-order,
neglected

DC Model

$$I_a = D \cdot I_c$$

$$V_{cp} = D \cdot (V_{ap} - I_c r_e D')$$

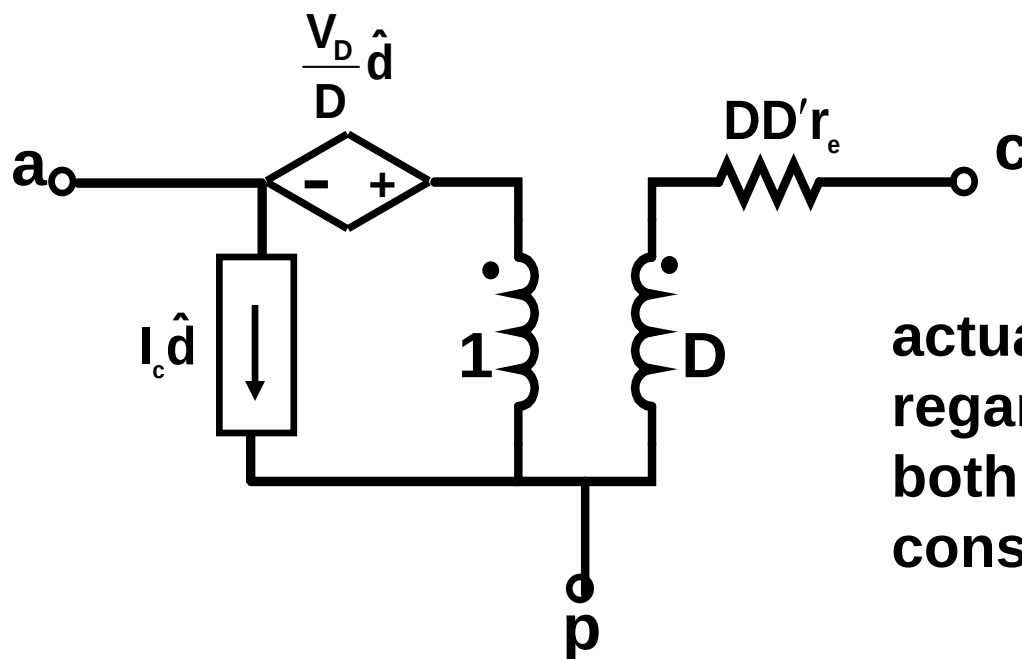


Small Signal Model

$$\hat{i}_a = D \cdot \hat{i}_c + I_c \cdot \hat{d}$$

$$\hat{v}_{cp} = D \cdot \hat{v}_{ap} + (V_{ap} + I_c D r_e - I_c D' r_e) \cdot \hat{d} - D D' r_e \cdot \hat{i}_c$$

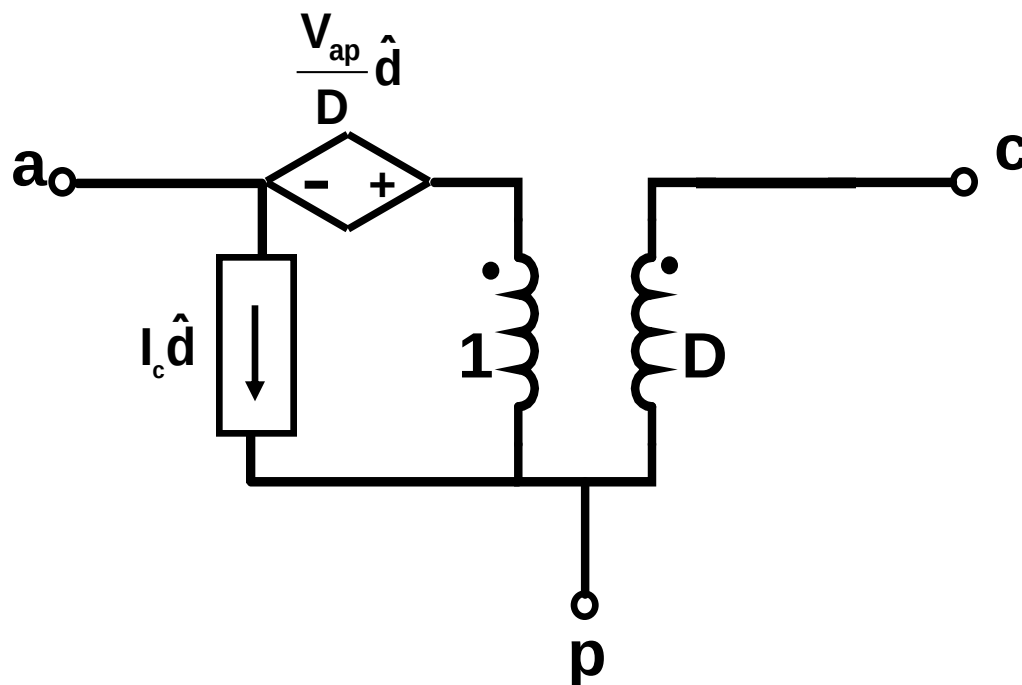
Define: $V_D = V_{ap} + I_c (D - D') r_e$



the $DD' r_e$ term only exist if the V_{ap} is not continuous, if V_{ap} is directly connect to voltage source, $DD' r_e$ will relinquish itself

actually this circuit can be regarded as the model for both DC and small signal, considering at DC: $\hat{d} = 0$

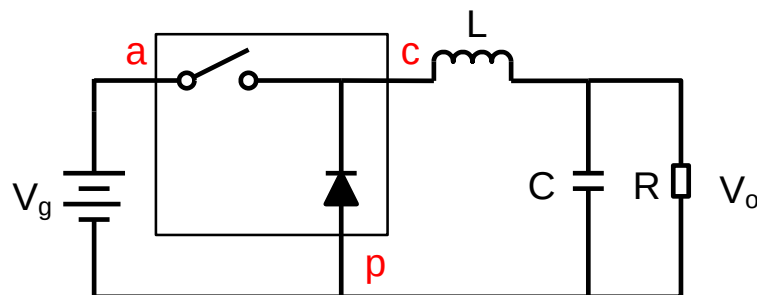
Under circumstances where r_e is negligible, previous model can be further simplified as:



Open-Loop Small-Signal Characteristics Using the Model of the PWM Switch

- **Line-to-Output Transfer Function**
- **Control-to-Output Transfer Function**
- **Output Impedance**

The Buck Converter



DC Analysis

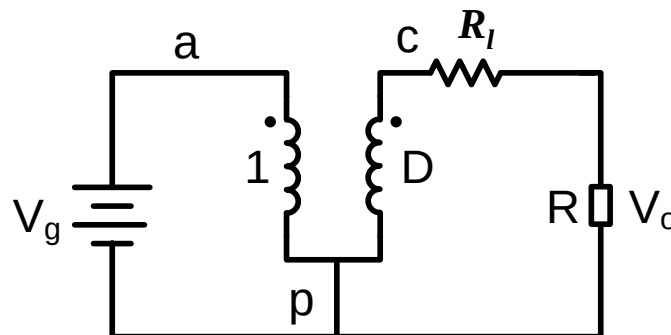
Under dc conditions, point-by-point by substitution of the PWM switch model results in the circuit shown below. The conversion ratio is easily determined:

$$M = \frac{V_o}{V_g} = D \frac{R}{R + R_l} = D \frac{1}{1 + \frac{R_l}{R}} \approx D \quad (1)$$

We also need the operating point (V_{ap} , I_c) which is required in the small-signal model:

$$V_{ap} = V_g \quad (2a)$$

$$I_c = \frac{V_o}{R} = \frac{DV_g}{R} \quad (2b)$$



Open-Loop Line-to-Output Transfer Function

In determining the open-loop line-to-output transfer function, **the duty-ratio controlled sources do not play any role and hence are removed.** The resultant circuit is show below whence we have:

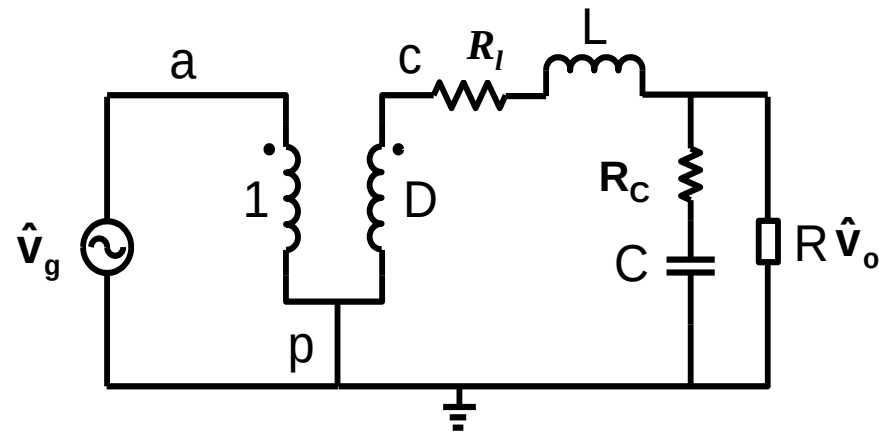
$$\mathbf{G}_{vg2vo}(s) = \frac{\hat{v}_o(s)}{\hat{v}_g(s)} = M \frac{\left(1 + \frac{s}{s_{z1}}\right)}{1 + \frac{s}{\omega_0 Q} + \frac{s^2}{\omega_0^2}} \quad (3)$$

where:

$$s_{z1} = \frac{1}{R_C C} \quad (4a)$$

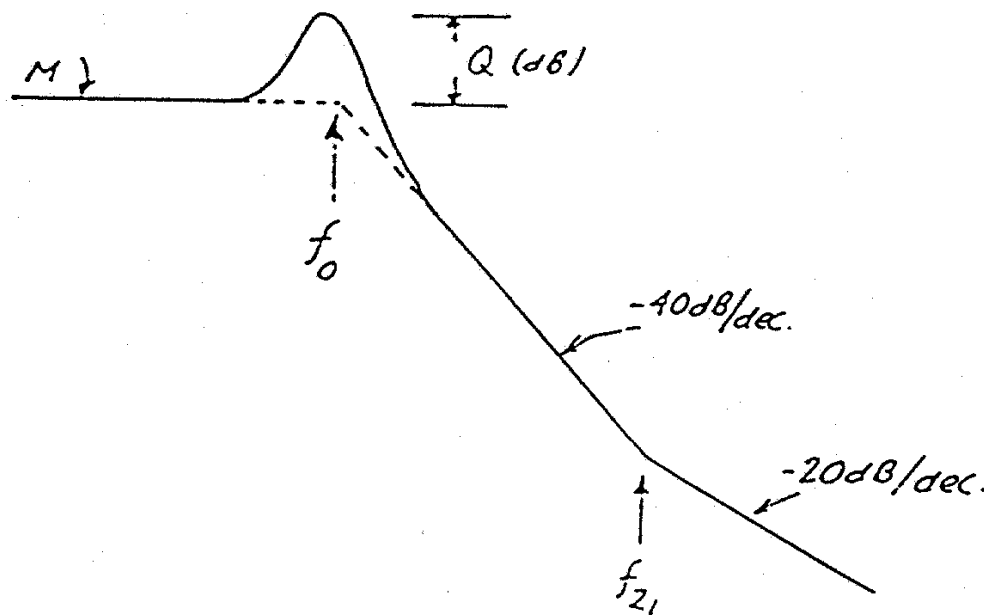
$$\omega_0 = \frac{1}{\sqrt{LC}} \sqrt{\frac{1 + \frac{R_l}{R}}{1 + \frac{R_C}{R}}} \approx \frac{1}{\sqrt{LC}} \quad (4b)$$

$$Q = \frac{1}{\omega_0} \frac{1}{\frac{L}{R_l + R} + C(R_C + R_l \parallel R)} \quad (4c)$$



Open-Loop Line-to-Output Transfer Function

The asymptotic magnitude Bode plot is shown below:



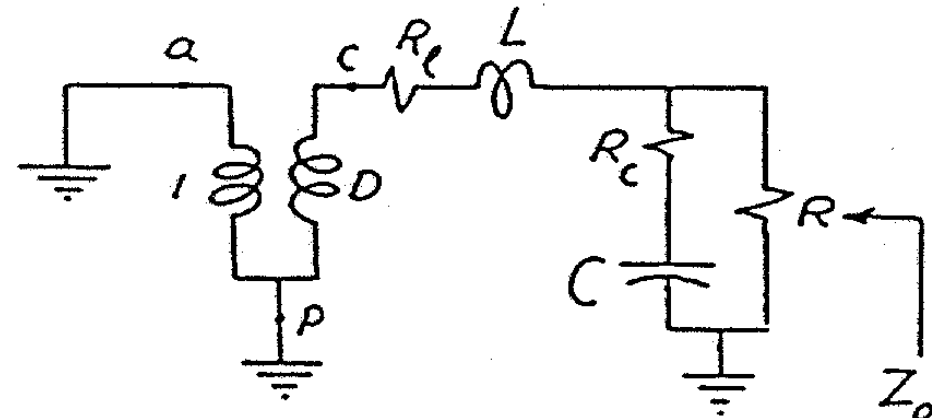
Open-Loop Output Impedance

The equivalent circuit for determining the output impedance is shown below.

Hence we have:
$$Z_O = \left(R_C + \frac{1}{sC} \right) \parallel (sL + R_l) \parallel R \quad (5)$$

which can be put in factored form as:

$$Z_O = R_l \parallel R \frac{\left(1 + \frac{s}{s_{z1}} \right) \left(1 + \frac{s}{s_{z2}} \right)}{1 + \frac{s}{\omega_0 Q} + \frac{s^2}{\omega_0^2}} \quad (6)$$

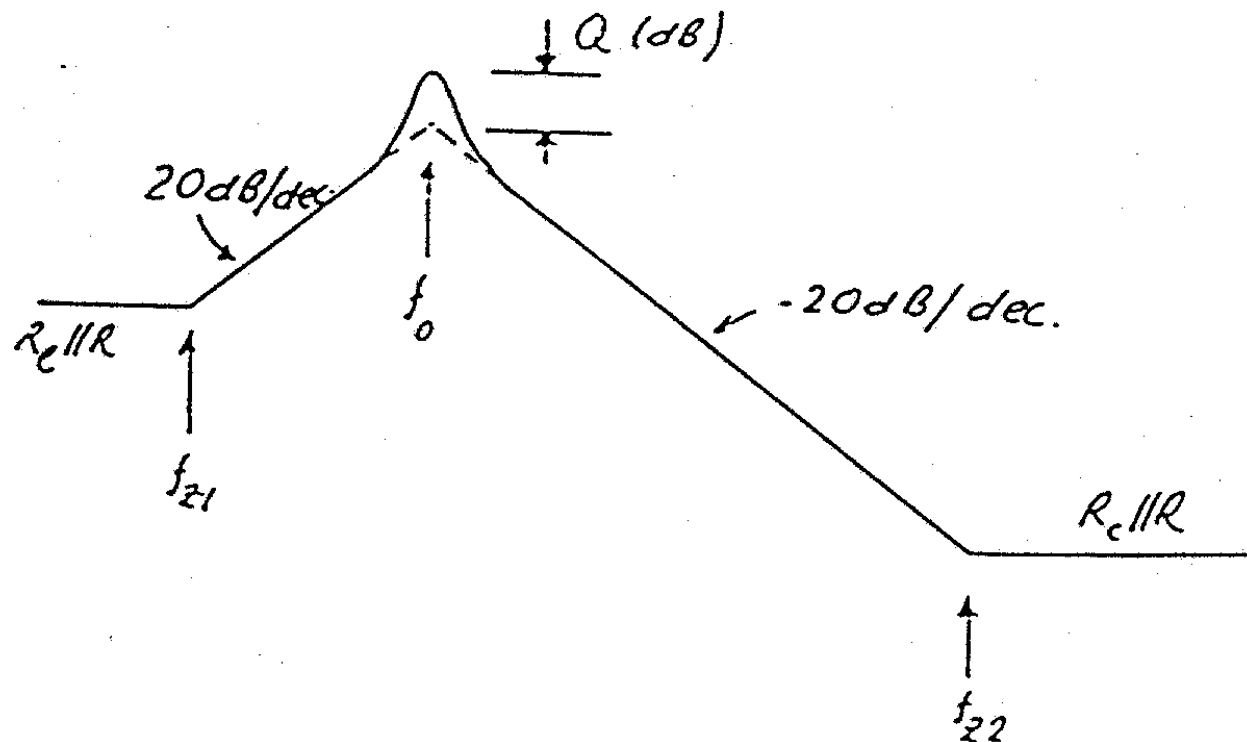


$$s_{z1} = \frac{R_l}{L}, \quad s_{z2} = \frac{1}{R_C C} \quad (7a-b)$$

ω_0 and Q are given in Eqs. (4b-c)

Open-Loop Output Impedance

The asymptotic magnitude Bode plot is shown below:



Open-Loop Control-to-Output Transfer Function CPES

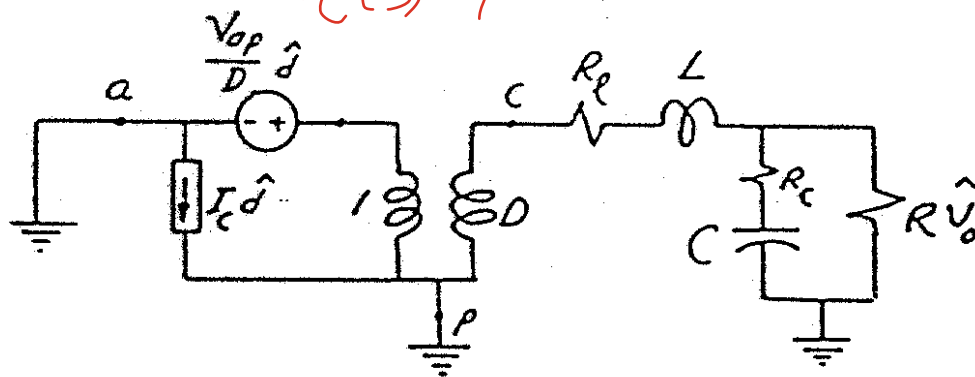
In this case $\hat{v}_g = 0$ and as a result the controlled current source is shunted as shown in the figure below. We have :

$$\frac{V_{ap}}{D} \hat{d}(s) \cdot H(s) = \hat{v}_o(s) \quad (8)$$

Where, $H(s) = G_{vg2vo}(s)$, as given in Eq. (3)

Using $V_{ap} = V_g$ as determined earlier from the dc analysis and the relation $\hat{d}(s) = \hat{v}_c(s) K_m$ for the modulator, we get :

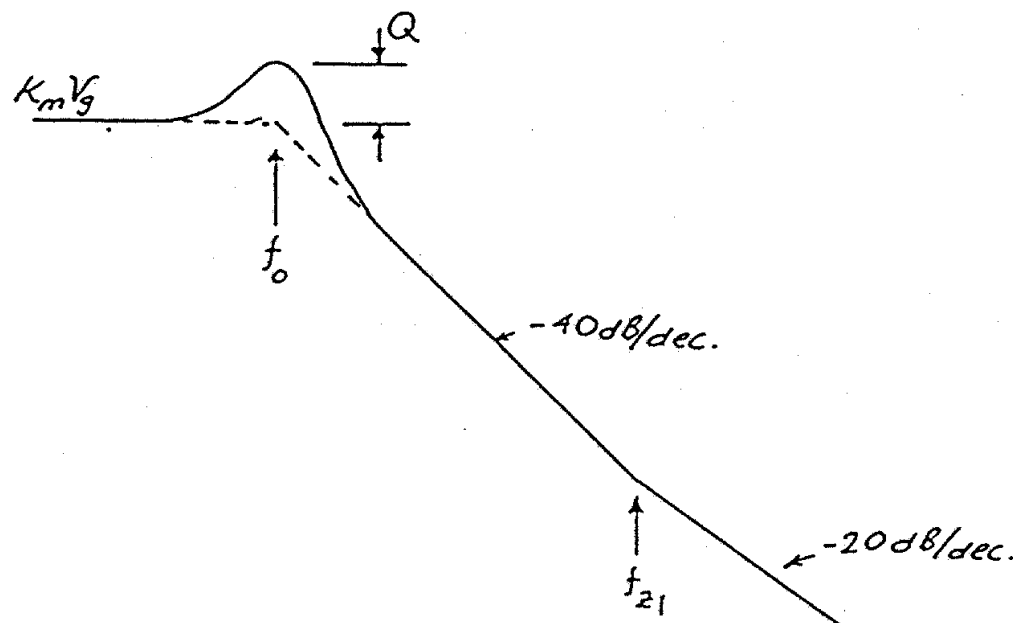
$$\frac{\hat{d}(s)}{\hat{v}_c(s)} = \frac{1}{V_p} \frac{\hat{v}_o(s)}{\hat{v}_c(s)} = K_m V_g \frac{1 + s/s_{z1}}{1 + s/\omega_0 Q + s^2/\omega_0^2} \quad (9)$$



where s_{z1} , ω_0 and Q are the same as those given earlier by Eqs. (4a - c).

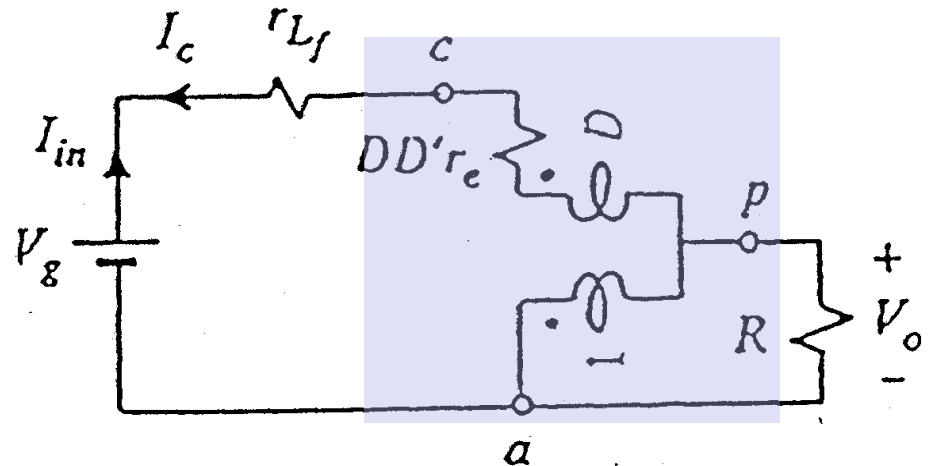
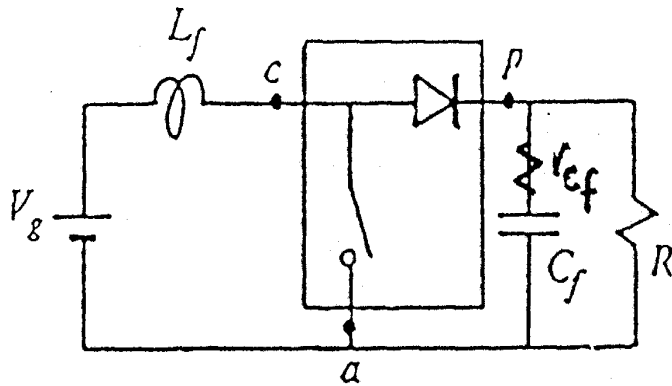
Open-Loop Control-to-Output Transfer Function

The asymptotic magnitude Bode plot is shown below:



The Boost Converter

- DC Analysis

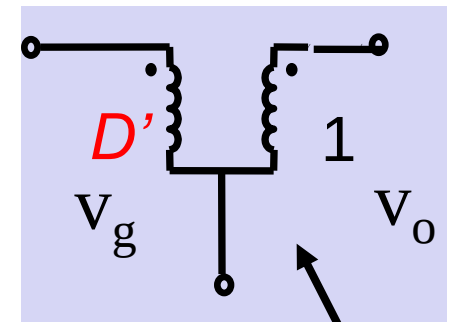


$$M = \frac{V_o}{V_g} = \frac{1}{D'} \frac{1}{1 + \frac{r_{L_f}}{D'^2 R} + \frac{r_e D}{R D'}}$$

$$V_{ap} = -V_o$$

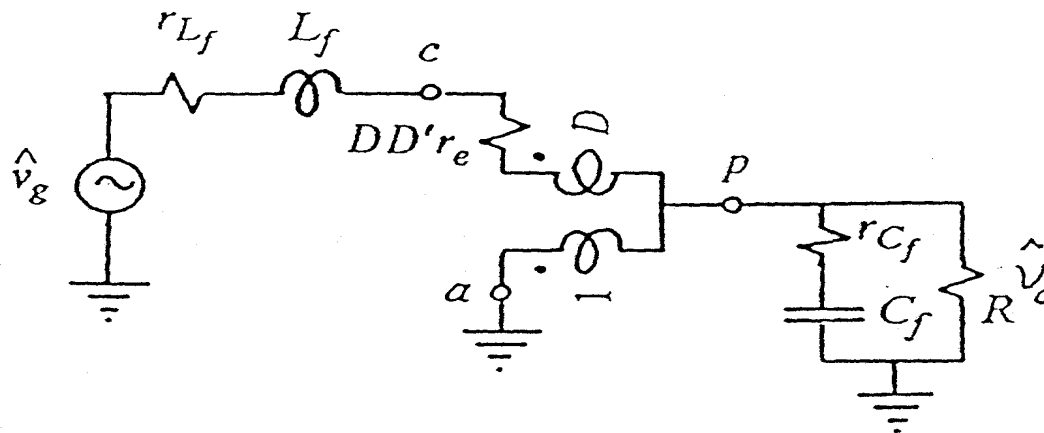
$$I_c = -I_{in} = -\frac{I_o}{D'}$$

$$r_e = r_{C_f} \parallel R$$



From chapter01

Line - to - Output $\hat{v}_o(s)/\hat{v}_g(s)$



$$\frac{\hat{v}_o(s)}{\hat{v}_g(s)} = M \frac{\left(1 + \frac{s}{s_{z1}}\right)}{1 + \frac{s}{\omega_0 Q} + \frac{s^2}{\omega_0^2}}$$

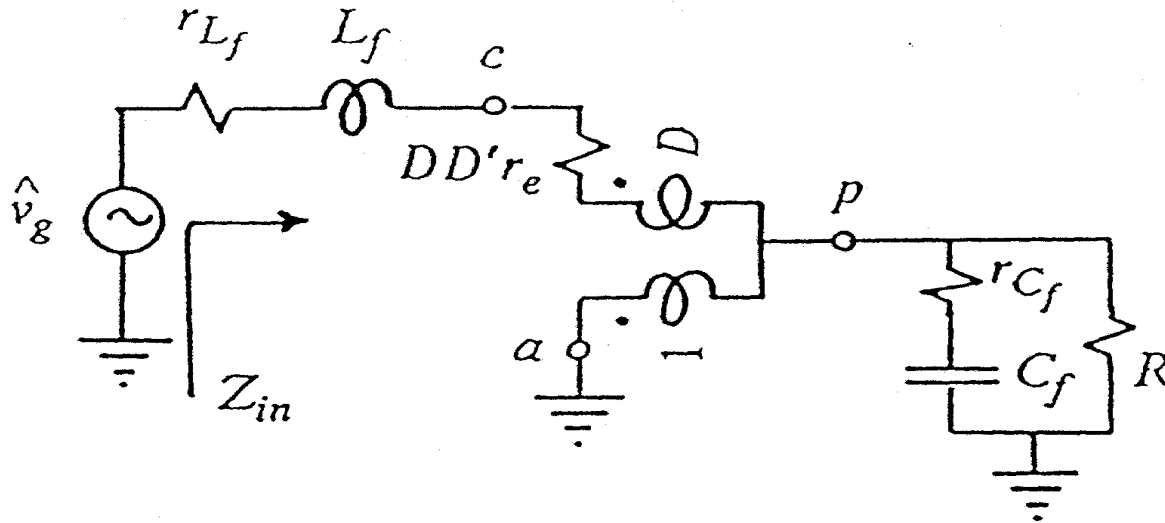
where:

$$Q = \frac{\omega_0}{\frac{r_{L_f} + r_e D'}{L_f} + \frac{1}{C_f(r_{C_f} + R)}}$$

$$\omega_0 = \frac{1}{\sqrt{L_f C_f}} \sqrt{\frac{r_{L_f} + r_e D D' + D'^2 R}{r_{C_f} + R}}$$

$$s_{z1} = \frac{1}{r_{C_f} C_f}$$

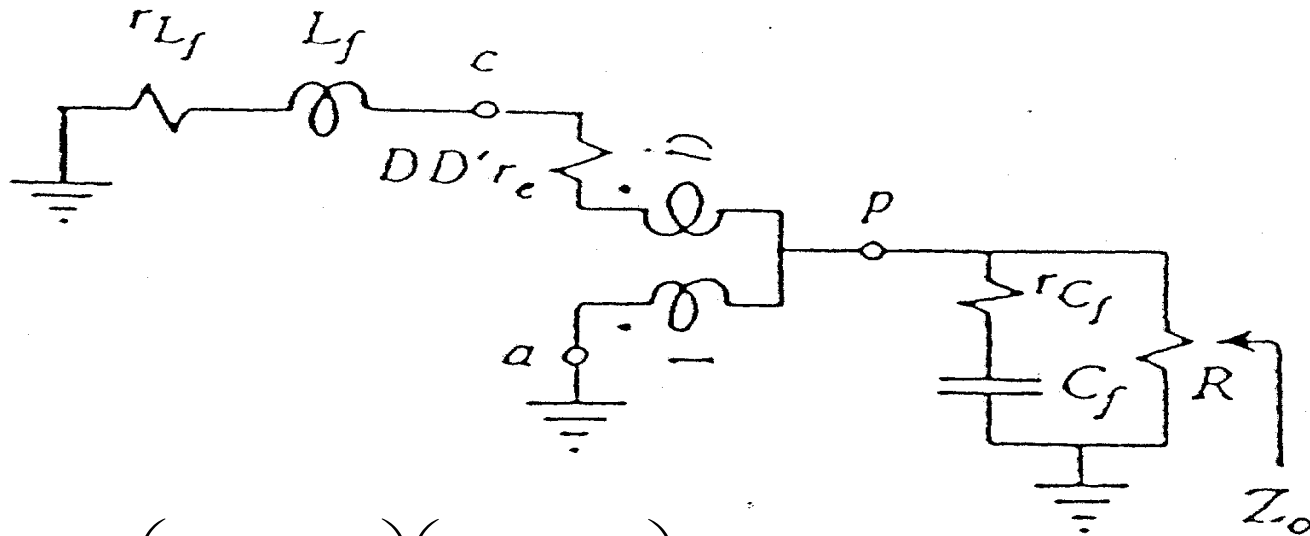
Input Impedance $Z_{in}(s)$



$$Z_{in} = R_{in} \frac{1 + \frac{s}{\omega_0 Q} + \frac{s^2}{\omega_0^2}}{1 + \frac{s}{s_p}}$$

$$R_{in} = r_{L_f} + r_e DD' + D'^2 R \qquad s_p = \frac{1}{C_f (r_{C_f} + R)}$$

Output Impedance $Z_o(s)$



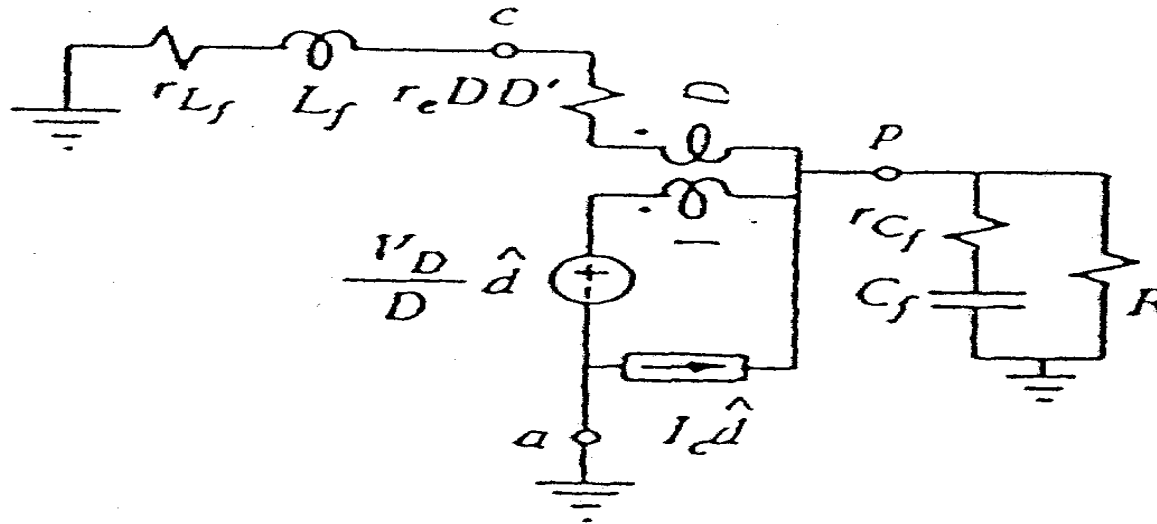
$$Z_o = R_o \frac{\left(1 + \frac{s}{s_{z0}}\right) \left(1 + \frac{s}{s_{z1}}\right)}{1 + \frac{s}{\omega_0 Q} + \frac{s^2}{\omega_0^2}}$$

$$R_o = R \parallel \frac{r_{L_f} + r_e DD'}{D'^2}$$

$$s_{z0} = \frac{r_{L_f} + r_e DD'}{L_f}$$

$$s_{z1} = \frac{1}{r_{C_f} C_f}$$

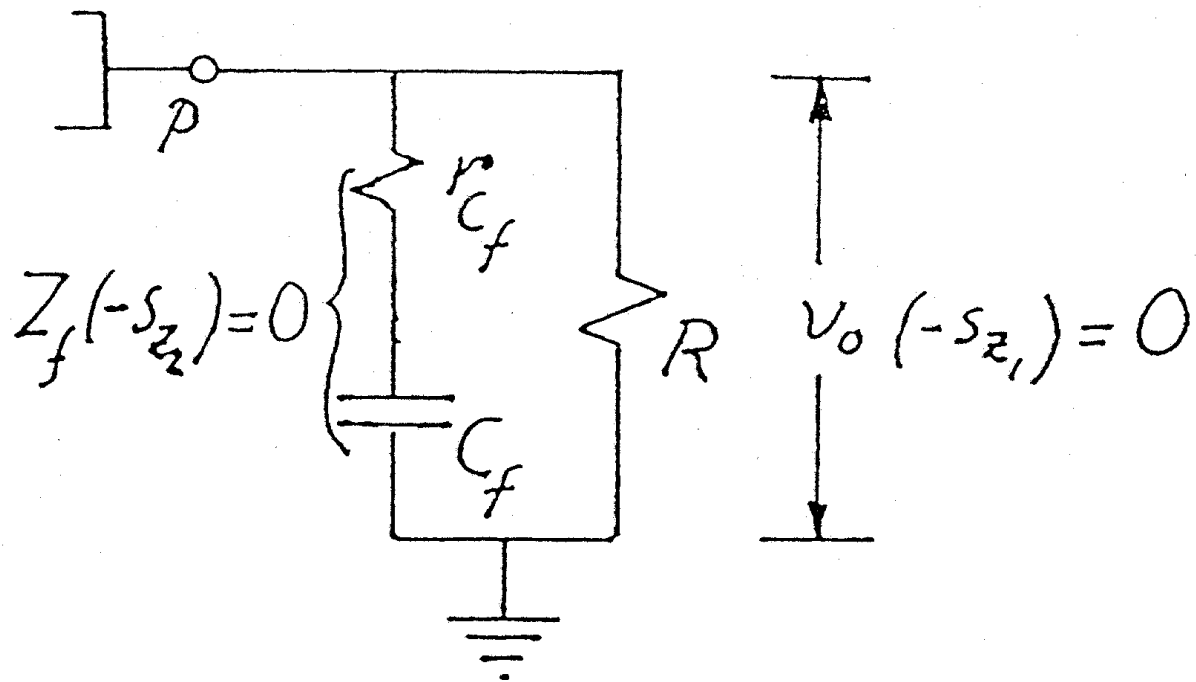
Control - Output $\hat{v}_o(s)/\hat{d}(s)$



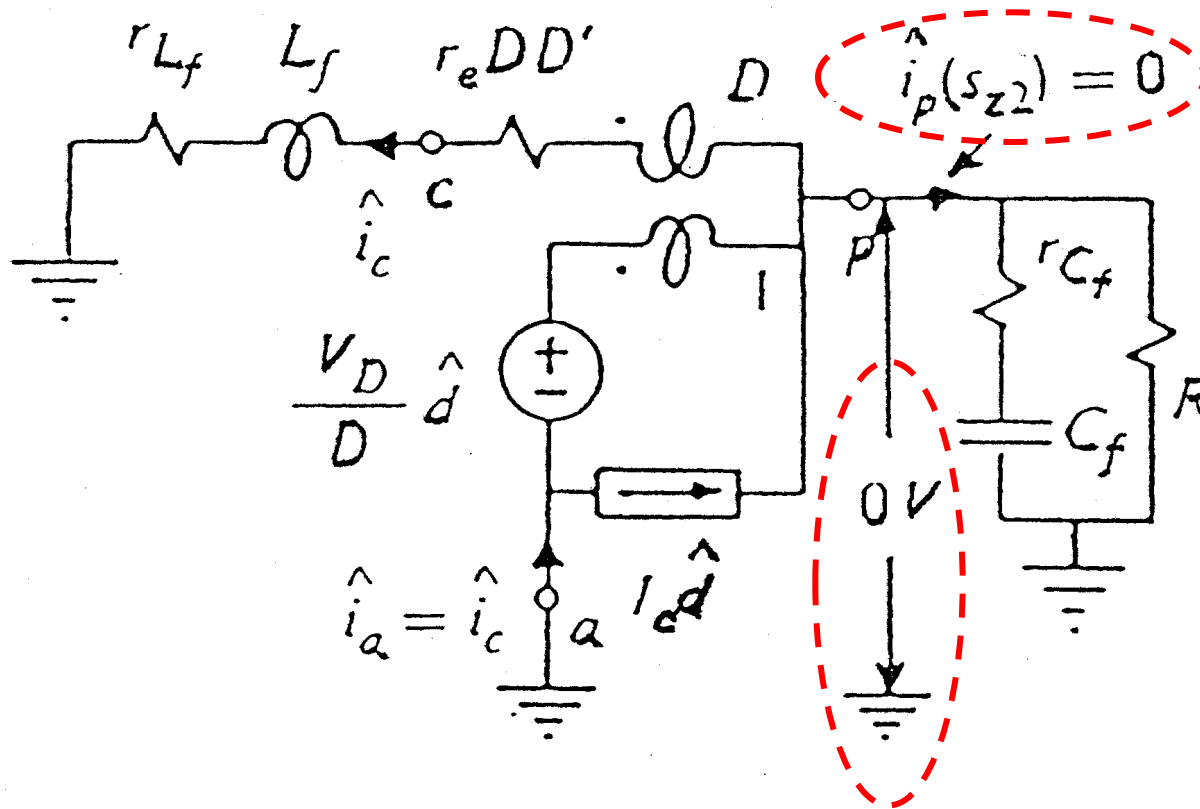
$$\frac{\hat{v}_o(s)}{\hat{d}(s)} = K_d \frac{\left(1 + \frac{s}{s_{z1}}\right) \left(1 - \frac{s}{s_{z2}}\right)}{1 + \frac{s}{\omega_0 Q} + \frac{s^2}{\omega_0^2}}$$

$$K_d = \frac{dV_o}{dD} = V_g \frac{dM}{dD} \approx \frac{V_g}{D'^2}$$

$$S_{z1} = ?$$



$$S_{z2} = ?$$



$$\mathbf{S}_{z2} = ?$$

Because $\hat{i}_p(s_{z2}) = 0$

We have: $\hat{i}_c = \hat{i}_a$

so that at the active terminal we have:

$$\hat{i}_c = I_c \hat{d} + \hat{i}_c D \rightarrow \hat{i}_c = \frac{I_c \hat{d}}{D'}$$

We also have: $\hat{\mathbf{i}}_c (\mathbf{R}_i + \mathbf{s}_{z2} \mathbf{L}_f) = \mathbf{V}_D \hat{\mathbf{d}}$

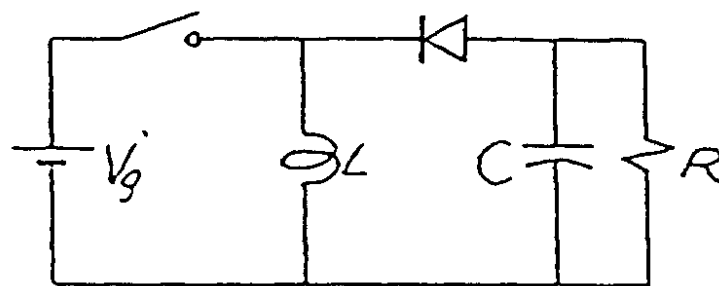
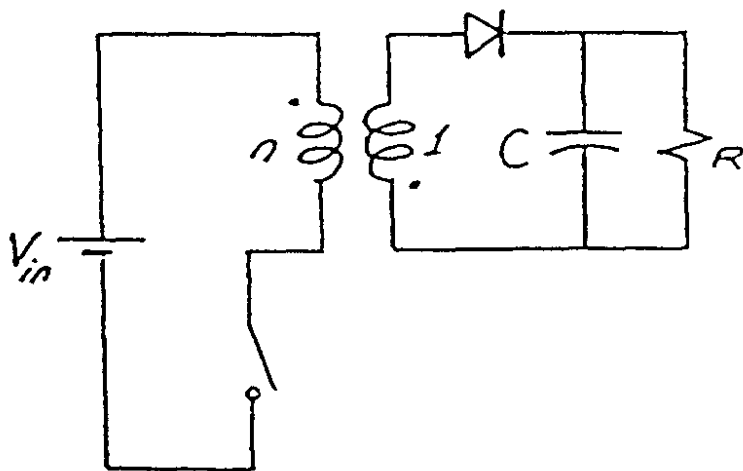
Where $\mathbf{R}_i = \mathbf{r}_{Lf} + \mathbf{D} \mathbf{D}' \mathbf{r}_e$ $\mathbf{V}_D = \mathbf{V}_{ap} + \mathbf{I}_c (\mathbf{D} - \mathbf{D}') \mathbf{r}_e$

$$\mathbf{V}_{ap} = -\mathbf{V}_O \quad \mathbf{I}_c = -\frac{\mathbf{I}_o}{\mathbf{D}'}$$

$$\mathbf{S}_{z2} = \mathbf{D}'^2 (\mathbf{R} - \mathbf{r}_e) / \mathbf{L}_f - (\mathbf{r}_{Lf} / \mathbf{L}_f) \approx \mathbf{D}'^2 \mathbf{R} / \mathbf{L}_f$$

The Flyback Converter

The Flyback converter is shown in the left figure. When the primary circuit is reflected into the secondary side, the buck-boost converter is obtained as shown in the right figure.



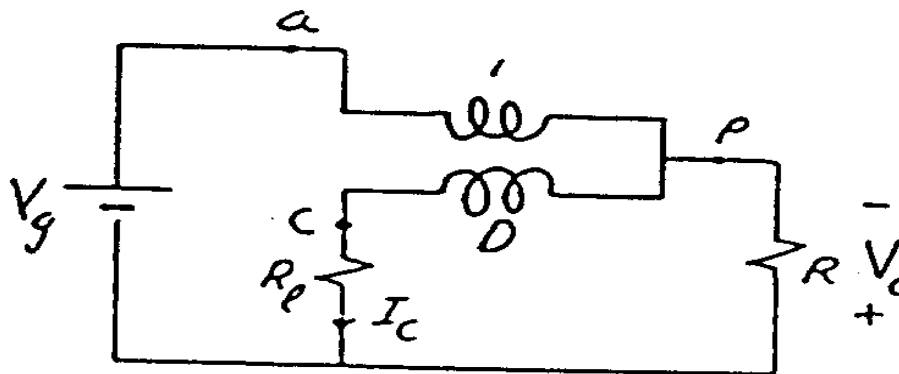
DC Analysis

Under dc conditions we have the circuit below from which we get:

$$M = \frac{V_O}{V_g} = \frac{D}{D'} \frac{1}{1 + \frac{R_L}{D'^2 R}} \quad (10)$$

We also have:

$$V_{ap} = V_g + V_O = V_g(1 + M) \quad I_C = I_g + I_O = \frac{MV_g}{RD'} \quad (11a-b)$$

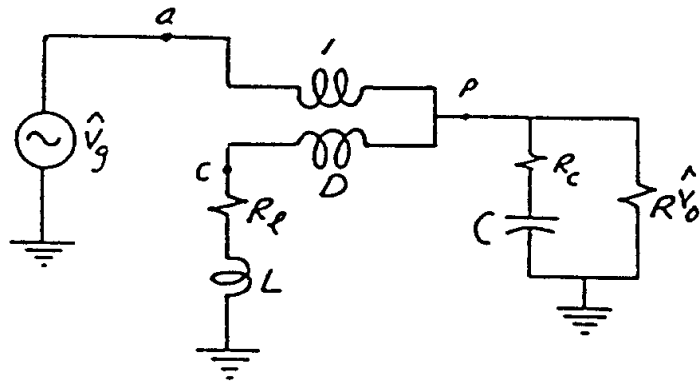


Open-Loop Line-to-Output Transfer Function

The equivalent circuit is shown below. We have:

$$\frac{\hat{v}_o(s)}{\hat{v}_g(s)} = M \frac{1 + \frac{s}{s_{z1}}}{1 + \frac{s}{\omega_0 Q} + \frac{s^2}{\omega_0^2}} \quad (12)$$

where: $s_{z1} = \frac{1}{R_c C}$ (13a)

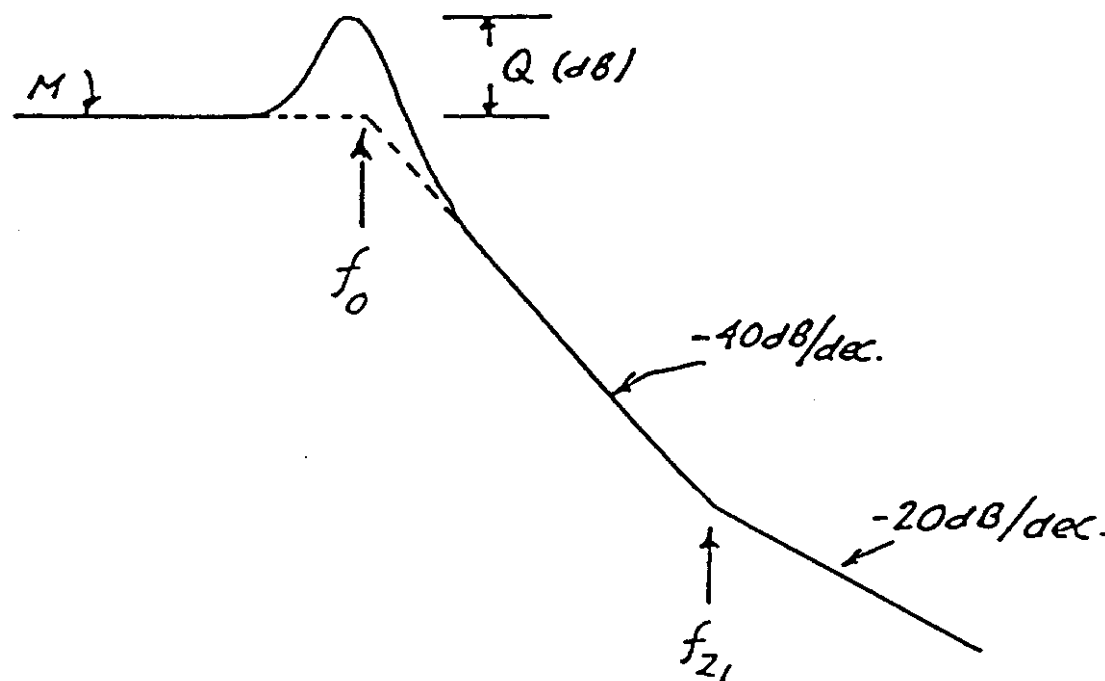


$$\omega_0 = \frac{1}{\sqrt{LC}} \sqrt{\frac{D'^2 + \frac{R_L}{R}}{1 + \frac{R_c}{R}}} \approx \frac{D'}{\sqrt{LC}} \quad (13b)$$

$$Q = \frac{1}{\omega_0} \frac{1}{\frac{L}{R_L + D'^2 R} + C \left[R_c + R \parallel \left(\frac{R_L}{D'^2} \right) \right]} \approx \frac{D' R}{\sqrt{L/C}} \quad (13c)$$

Notice that the approximations given here are the same as those obtained earlier using the method of state-space analysis. Also note the slight discrepancy between the exact expressions of ω_0 and Q obtained here using the switch model and those obtained using the method of state-space analysis.

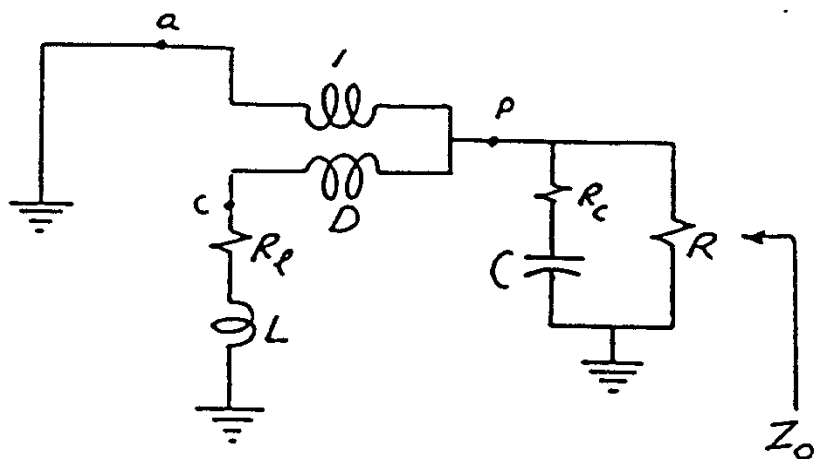
The asymptotic magnitude Bode plot is shown below:



Open-Loop Output Impedance

From the equivalent circuit shown below we have:

$$Z_O = \left(\frac{R_L}{D'^2} \right) \parallel R \frac{(1 + s/s_{z1})(1 + s/s_{z2})}{1 + s/\omega_0 Q + s^2/\omega_0^2} \quad (14)$$



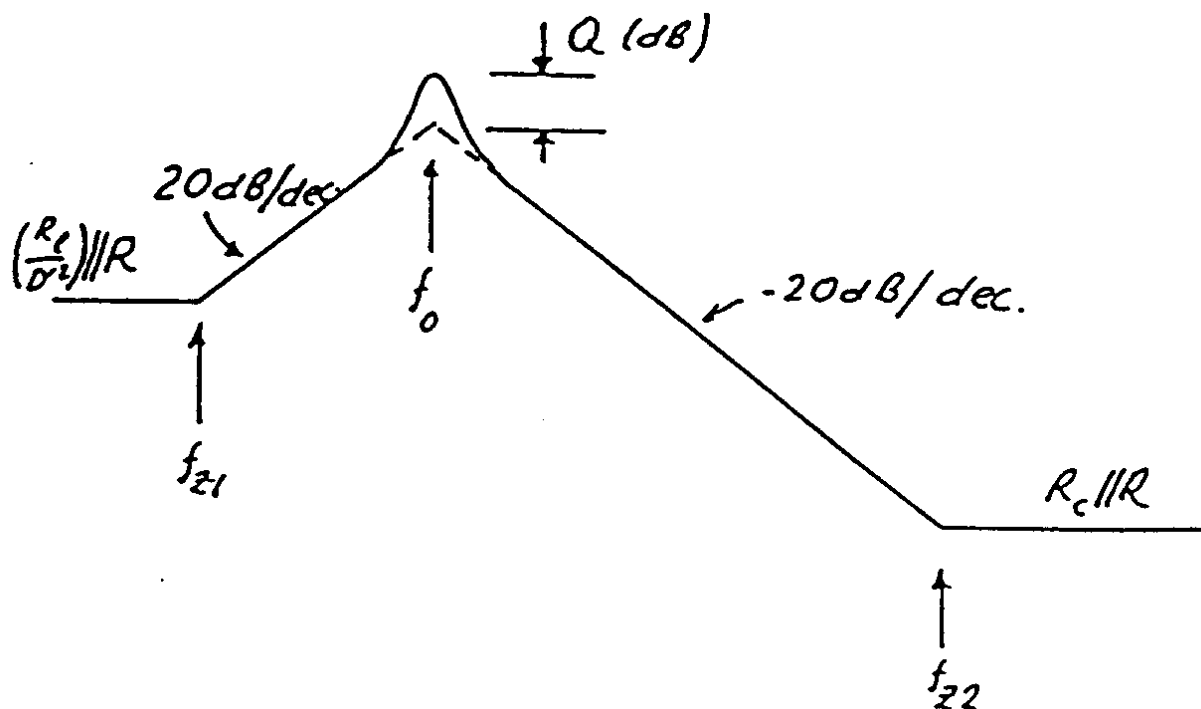
Where ω_0 and Q are given Eqs.(13b-c) and:

$$s_{z1} = \frac{R_L}{L} \quad (15a)$$

$$s_{z2} = \frac{1}{R_C C} \quad (15b)$$

Open-Loop Output Impedance

The asymptotic magnitude Bode plot is shown below:



Open-Loop Control-to-Output Transfer Function

From the equivalent circuit below we have:

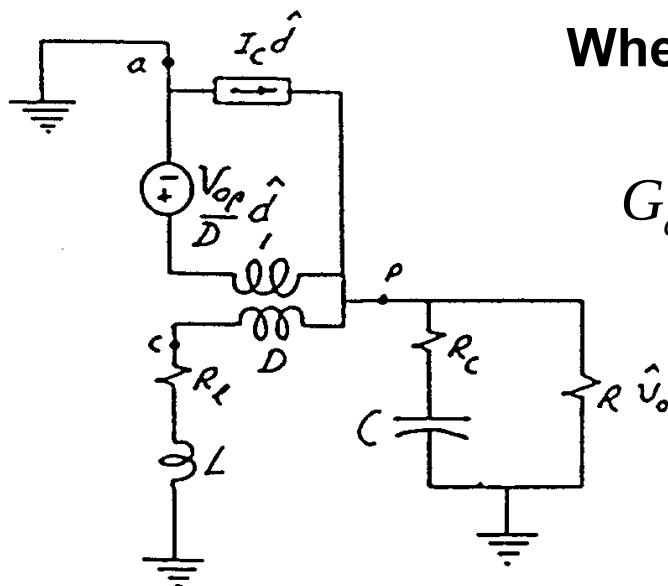
$$\frac{\hat{v}_o(s)}{\hat{v}_c(s)} = K_m G_{do} \frac{(1 + s/s_{z1})(1 - s/s_{z2})}{1 + s/\omega_0 Q + s^2/\omega_0^2} \quad (16)$$

Where ω_0 and Q are given in Eqs. (13b-c) and:

$$G_{do} = V_g \frac{D'^2 + (D' - D)R_L/R}{(D'^2 + R_L/R)^2} \approx \frac{V_g}{D'^2} \quad (17a)$$

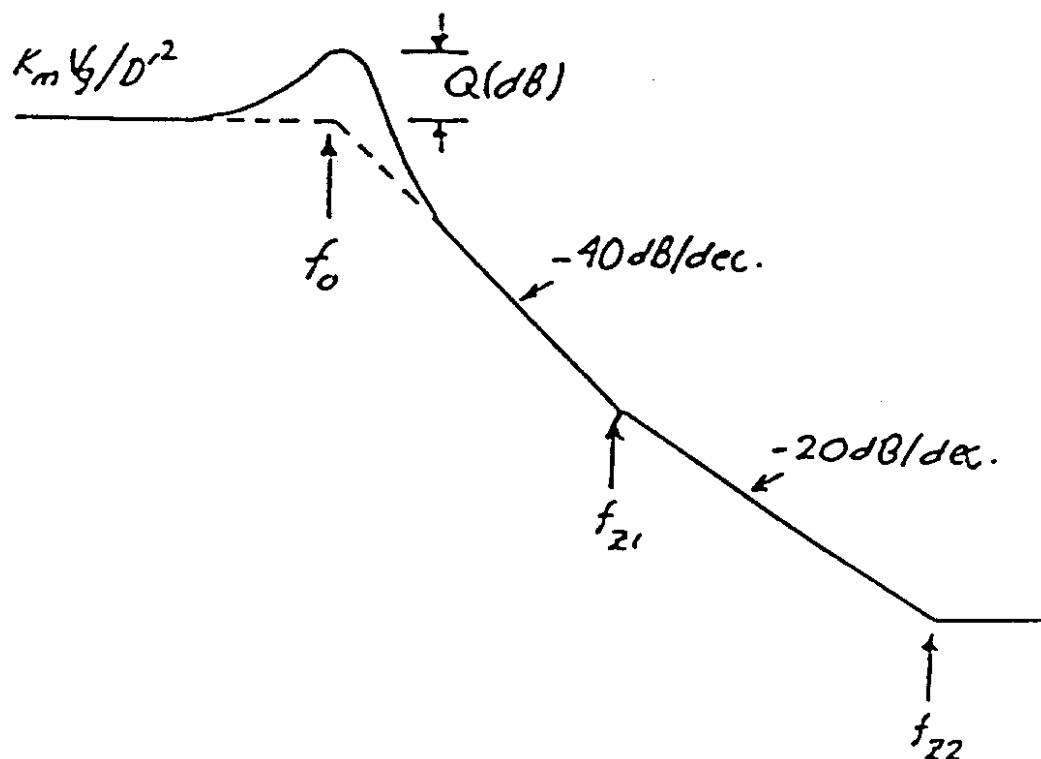
$$s_{z1} = \frac{1}{R_c C} \quad (17b)$$

$$s_{z2} = \frac{D'^2 R + R_1(D' - D)}{DL} \approx \frac{D'^2 R}{DL} \quad (17c)$$

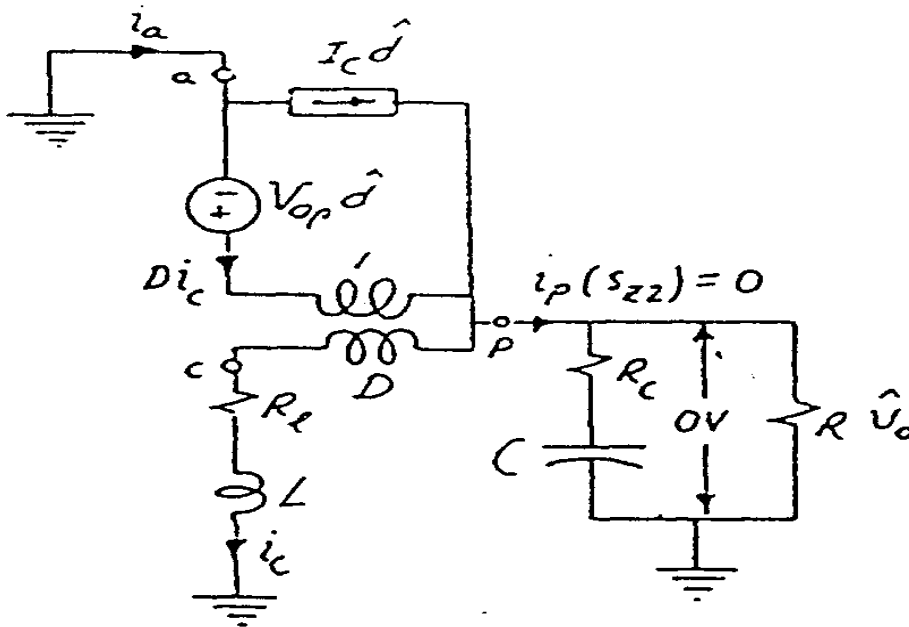


Open-Loop Control-to-Output Transfer Function

The asymptotic magnitude Bode plot is shown below:



Right-Half-Plane Zero



$$i_c = i_a$$

$$i_c = I_c \hat{d} + i_c D \rightarrow i_c = \frac{I_c \hat{d}}{D'}$$

$$i_c (R_L + s_{z2} L) \frac{1}{D} = \frac{V_{ap}}{D} \hat{d}$$

$$s_{z2} = (D' \frac{V_{ap}}{I_c} - R_L) / L$$

$$s_{z2} = (D' \frac{V_{ap}}{I_c} - R_L) / L$$

$$V_{ap} = V_g + V_o = V_g(1 + M) \quad I_c = I_g + I_o = \frac{MV_g}{RD'}$$

$$M = \frac{V_o}{V_g} = \frac{D}{D'} \frac{1}{1 + \frac{R_i}{D'^2 R}}$$



$$s_{z2} = \frac{D'^2 R + R_i(D' - D)}{DL} \approx \frac{D'^2 R}{DL}$$

Chapter 2.2

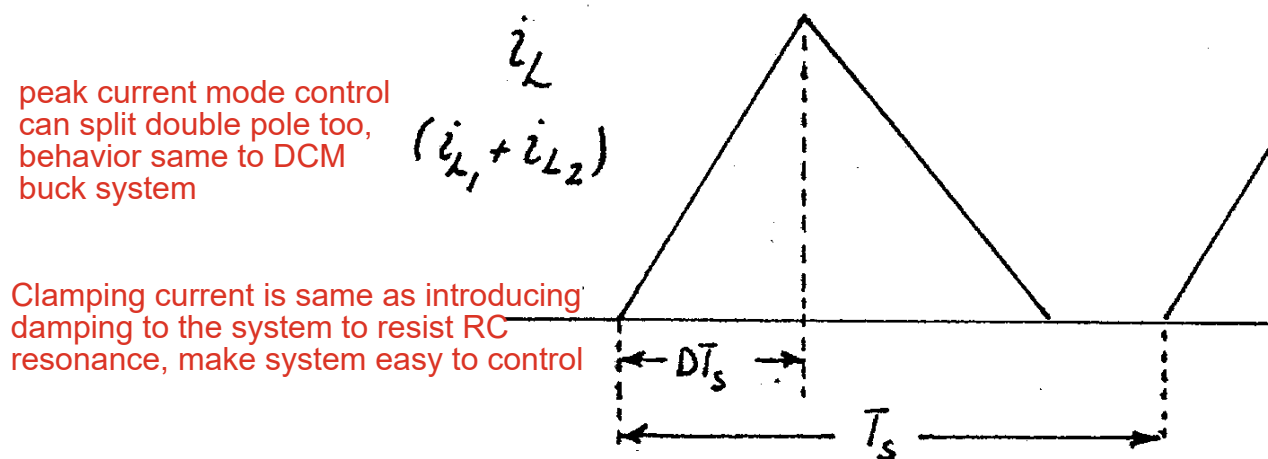
DISCONTINUOUS CONDUCTION MODE [2]

[2] Vatche Vorperian, "Simplified analysis of PWM converters using model of PWM Switch Part 2: Discontinuous Conduction Mode", IEEE Transactions on Aerospace and Electronic System, Vol.26, No.3 May 1990, pp. 497-505.

INTRODUCTION

In discontinuous conduction mode, PWM converters go through three switched networks during a switching cycle.

The inductor current in DCM starts from zero and terminates at zero before the end of the switching cycle. In the case of the Cuk and Cuk-derived converters the sum of the two inductor currents is constrained to be zero.



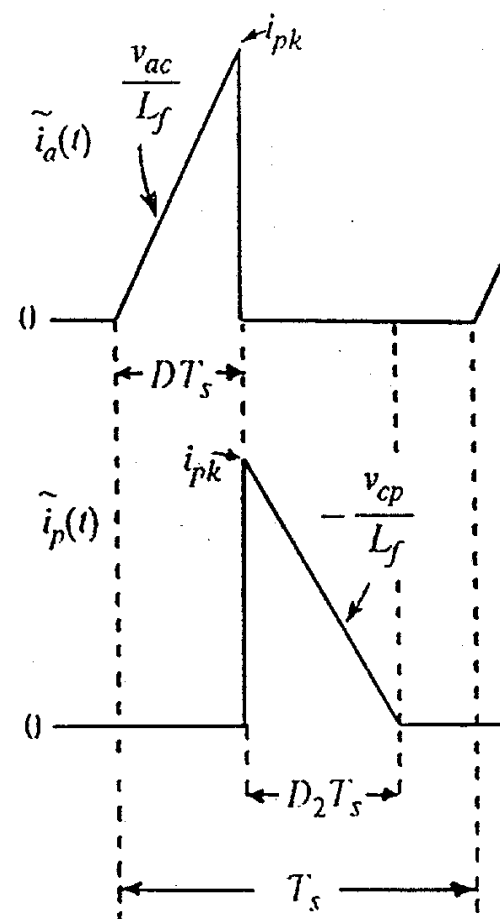
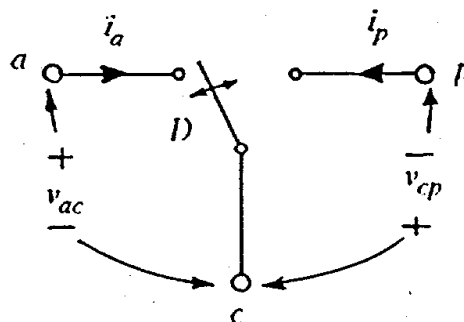
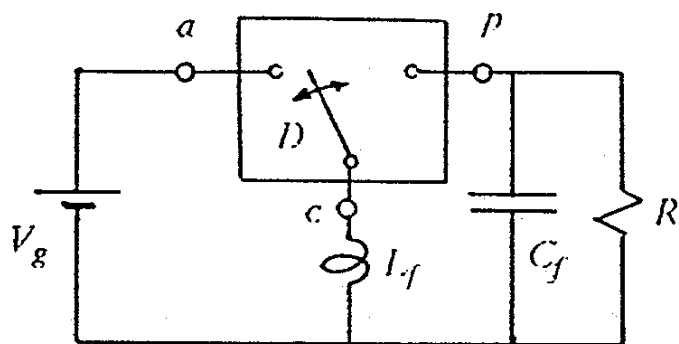
According to the instantaneous terminal currents shown, the following expressions for the **average** terminal currents and voltages can be easily verified:

$$i_a = \frac{i_{pk}}{2} d \quad (1)$$

$$i_p = \frac{i_{pk}}{2} d_2 \quad (2)$$

$$v_{ac} = L \frac{di_{pk}}{dT_s} \quad (3)$$

$$v_{cp} = L \frac{di_{pk}}{d_2 T_s} \quad (4)$$



From these equations the following relationships between the average terminal voltages and currents can be deduced:

$$i_a = \frac{d}{d_2} i_p \quad (5)$$

$$v_{ac} = \frac{d_2}{d} v_{cp} \quad (6)$$

There is no ripple information in the controller as only average value are taken as control value (Eq 5, Eq 6), the peak current are basically cancelled in the math procedure

$$d_2 = \frac{2LF_s}{d} \frac{i_a}{v_{cp}} = \frac{2LF_s}{d} \frac{i_p}{v_{ac}} \quad (7)$$

The average model follows from Eqs. (5-7) as show below where

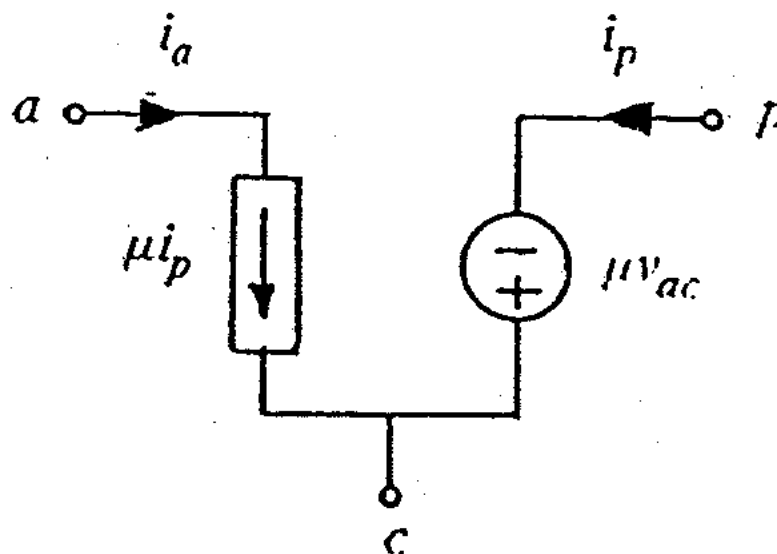
$$v_{cp} = \mu v_{ac} \quad (8a)$$

$$i_a = \mu i_p \quad (8b)$$

$$\mu = \frac{d^2}{2LF_s} \frac{v_{cp}}{i_a} = \frac{d^2}{2LF_s} \frac{v_{ac}}{i_p} \quad (8c)$$

When the average model is used in the dc analysis of PWM converters in DCM, the quantities in Eq.(8) are replaced by their dc values, i.e., D , V_{pc} , I_a , etc..

$$i_a = \mu i_p \quad V_{cp} = \mu V_{ac}$$



The average model of the PWM switch in DCM

The small-signal model is obtained from the relationship among the perturbation in average terminal quantities at a given dc operating point (I_p , V_{ac} , D).

Hence, perturbation of Eqs.(5) and (7) results in (after some algebra)

$$\hat{i}_a = \hat{v}_{ac} g_i + k_i \hat{d} \quad (9a)$$

where

$$g_i = \frac{I_a}{V_{ac}} \quad (9b)$$

$$k_i = \frac{2I_a}{D} \quad (9c)$$

Perturbation of Eqs.(6) and (7) results in

$$\hat{i}_p = g_f \hat{v}_{ac} + k_0 \hat{d} - g_0 \hat{v}_{cp} \quad (10a)$$

where

$$k_0 = \frac{2I_p}{D} \quad (10b)$$

$$g_0 = \frac{I_p}{V_{cp}} \quad (10c)$$

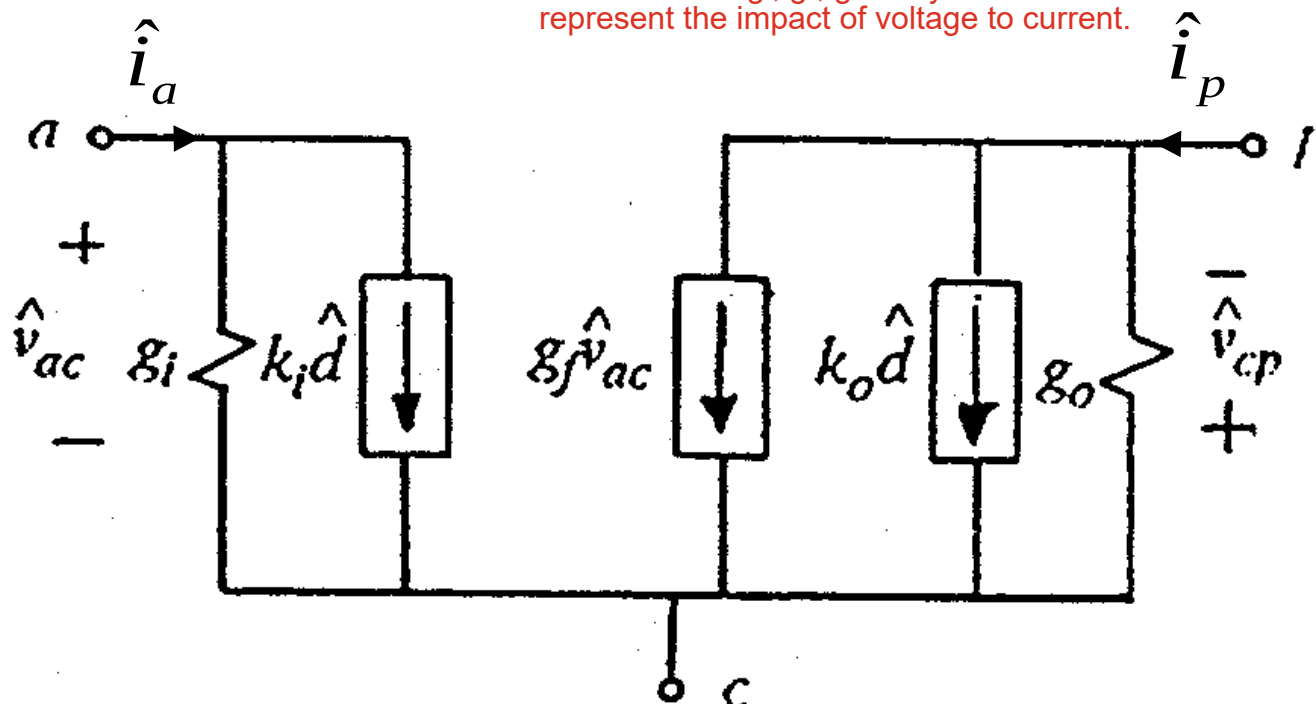
$$g_f = \frac{2I_p}{V_{ac}} \quad (10d)$$

Equations (9a) and (10a) correspond to the equivalent circuit model shown.

$$\hat{i}_a = \hat{v}_{ac} g_i + k_i \hat{d} \quad (9a)$$

$$\hat{i}_p = g_f \hat{v}_{ac} + k_o \hat{d} - g_o \hat{v}_{cp} \quad (10a)$$

virtual resistor g_i, g_f, g_o in system to represent the impact of voltage to current.



The small-signal model of the PWM switch in DCM

Model of the PWM Switch:

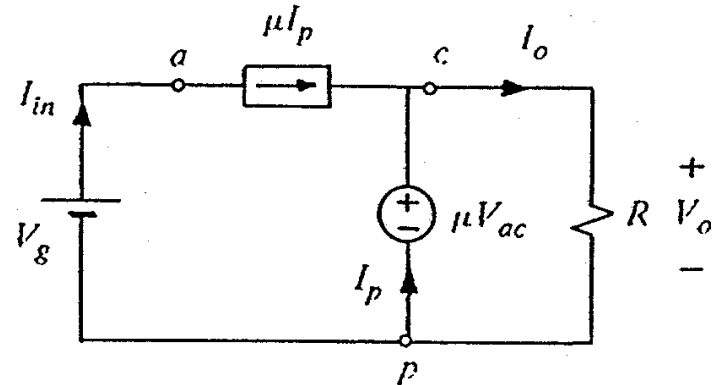
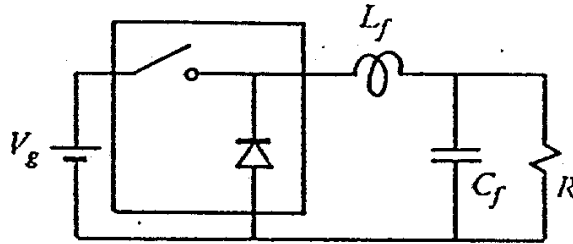
According to the model of the PWM switch in DCM, there is no reduction in the order of the average model of a PWM converter as it enters DCM.

The model of the PWM switch in DCM, in contrast to the model in CCM, has **small-signal resistors** which damp the low-pass filter to the point where two real poles are created. The first pole agrees with the pole of state-space averaging while the second pole occurs in the range $fp2 > F_s/\pi$.

The small-signal resistances in the model of the switch in DCM represent the load dependent nature of PWM converters in DCM.

Analysis of Buck Converters

Using the Model of the PWM Switch in DCM



DC Analysis

Under dc conditions, point-by-point substitution of the average PWM switch model in the buck converter results in the circuit shown from which we get:

$$V_O = \mu V_{ac} = \mu(V_g - V_O) \quad (11)$$

which gives

$$M = \frac{V_O}{V_g} = \frac{1}{1 + 1/\mu} \quad (12)$$

The amplification factor, μ , of the switch is determined in terms of the converter quantities as follows:

$$\mu = \frac{D^2}{2LF_s} \frac{V_{cp}}{I_a} = \frac{D^2}{2LF_s} \frac{V_o}{I_{in}} = \frac{D^2}{KM} \quad (13)$$

where

$$M = \frac{I_{in}}{I_o}, \quad K = \frac{2LF_s}{R} \quad (14)$$

Substitution of Eq.(14) in (12) gives

$$M = \frac{1}{1 + \frac{KM}{D^2}} \quad (15)$$

which is a quadratic equation in M the root of which is given by:

$$M = \frac{2}{1 + \sqrt{1 + 4K/D^2}} \quad (16)$$

The critical value of K which determines the boundary between DCM and CCM can be obtained from Eq.(15) by letting $M=D$, which is the value of M in CCM, and obtain:

$$K_{crit} = 1 - D \quad (17)$$

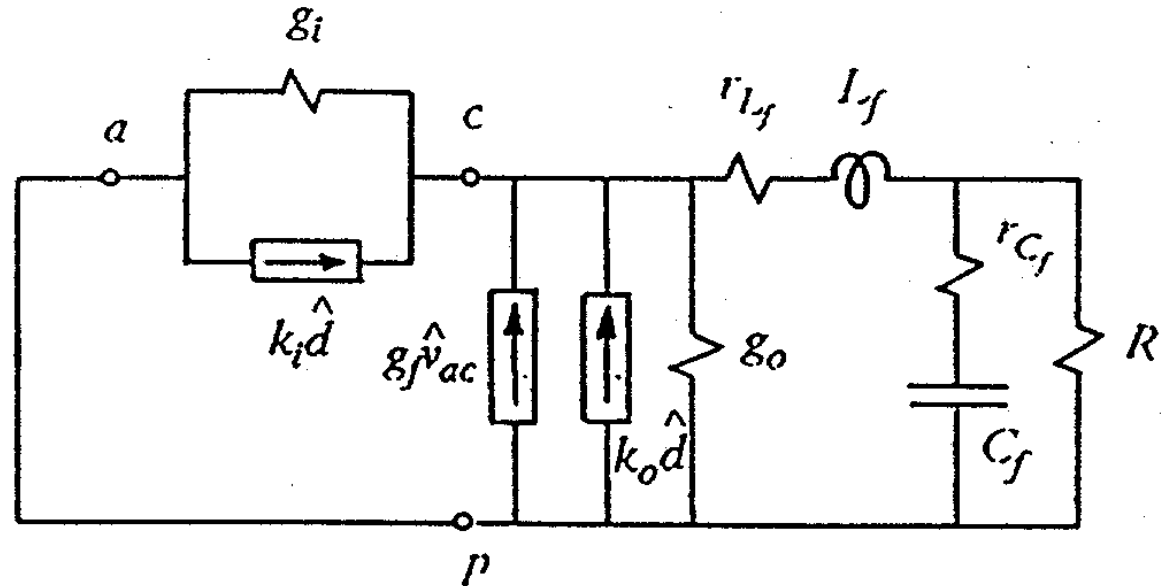
This completes the dc analysis of the buck in DCM.

Control-to-Output Transfer Function

To perform this analysis, we substitute the model of the PWM switch in DCM in the buck converter and we **short V_g** as shown. The small-signal current source $g_f v_{ac}$ connected between terminal *a* and *c* acts like a conductance g_f as can be easily verified. Hence, the three conductances g_o , g_i , and g_f can be combined into a single conductance while the two controlled current sources can be combined in a single current source as shown where

$$r = \frac{1}{g_i + g_o + g_f} \quad (18a)$$

$$k_d = k_i + k_o \quad (18b)$$

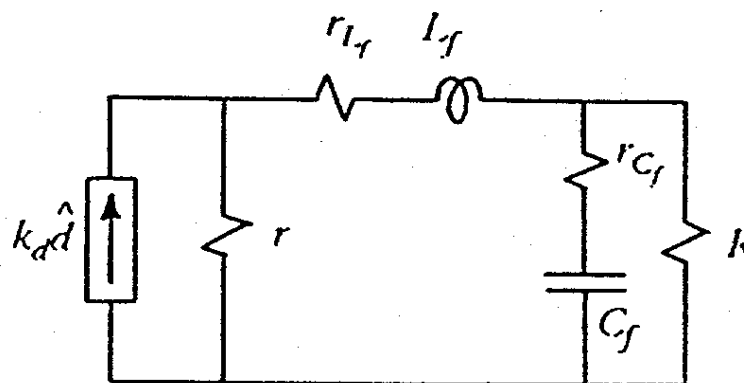


Next, we evaluate the small-signal parameters **at the dc operating point** using Eqs.(9c) and (10b) as follows:

$$k_i + k_o = \frac{2}{D} (I_a + I_p) = \frac{2I_o}{D} \quad (19)$$

Likewise, for r we obtain after a few lines of algebra and assumming r_{Lf} is negligible in the dc analysis:

$$r = \frac{1}{g_i + g_o + g_f} = R(1 - M) \quad (20)$$



The low freq pole will be controlled by capacitor; the high freq pole will be controlled by inductor (assuming idea current source case, L is infinite large)

when r is small, L and C are equivalent parallel with each other, the damping is samll, then LC will resonant; if r is big, L is a part of current source, so no resonance.

Then we will have two separte poles instead of a double pole, which is desired for controller design as it behaves like a single pole system, easy to be compensated.

The control-to-output transfer function is now obtained

$$\frac{\hat{v}_o}{\hat{d}} = H_d \frac{1 + s/s_{z1}}{1 + a_1 s + a_2 s^2} \quad (21)$$

where

$$H_d = \frac{2I_o}{D} \frac{rR}{r + R + r_{L_f}} \quad (22a)$$

$$s_{z1} = \frac{1}{r_{C_f} C_f} \quad (22b)$$

$$a_1 = \frac{L}{r_{L_f} + r + R} + C_f [r_{C_f} + R \parallel (r + r_{L_f})] \quad (22c)$$

$$a_2 = LC_f \frac{r_{C_f} + R}{r_{L_f} + r + R} \quad (22d)$$

Although at this point the control-to-output transfer function is completely determined, we should perform some analytical simplifications in order to see if familiar results known from state-space averaging are present in Eq.(22). To do so, we rewrite a_1 assuming $r_{Cf} = r_{Lf} = 0$ and obtain

$$a_1 = C_f R \frac{1-M}{2-M} \left[1 + \frac{1}{RC_f} \frac{L}{R} \frac{1}{1-M} \right]$$

The second term in the brackets can be shown to be much less than unity by writing it as follows

$$\frac{1}{RC_f} \frac{L}{R} \frac{1}{1-M} = \frac{T_s}{2RC_f} \frac{K}{1-M} \quad (23) \quad K = \frac{2LF_s}{R}$$

But, from Eq.(15) we have

$$\frac{1-M}{K} = \left(\frac{M}{D} \right)^2 \quad (24)$$

so that Eq.(23) can be written as

$$\frac{1}{RC_f} \frac{L}{R} \frac{1}{1-M} = \frac{T_s}{2RC_f} \left(\frac{D}{M} \right)^2 \ll 1 \quad (25)$$

The previous inequality is true because of two reasons. First, the switching time is much shorter than the output filter time constant and second, $D < M$ in DCM. Hence, we have

$$a_1 \cong RC_f \frac{1-M}{2-M} \quad (26) \quad a_2 \cong \frac{LC_f}{2-M} \quad \frac{a_2}{a_1} \cong \frac{L}{R(1-M)}$$

Using the inequality in Eq.(25), we can show that

$$a_1 \gg \frac{a_2}{a_1}$$

so that the quadratic can be factored as

$$1 + a_1 s + a_2 s^2 \approx (1 + a_1 s) \left(1 + \frac{a_2}{a_1} s \right)$$

Hence, the control-to-output transfer function can be written as:

$$\frac{\hat{V}_o}{\hat{d}} = H_d \frac{1 + s/s_{z1}}{(1 + s/s_{p1})(1 + s/s_{p2})} \quad (27)$$

where

$$s_{p1} = \frac{1}{a_1} = \frac{1}{RC} \frac{2-M}{1-M} \quad (27a)$$

$$s_{p2} = \frac{a_1}{a_2} = 2F_s \left(\frac{M}{D} \right)^2 \quad (27b)$$

It can be seen now that the dominant pole is given by Eq.(27a) which is the same as the single pole given by state-averaging. There is a second pole, however, which satisfies the following general equality:

$$f_{p2} = \frac{s_{p2}}{2\pi} \geq \frac{F_s}{\pi} \text{ Hz} \quad (27c)$$

which can be easily verified from Eq.(27b), as explained earlier, by noting that the conversion ration M in DCM is always larger than the conversion ratio in CCM, which for the buck converter is given by D .

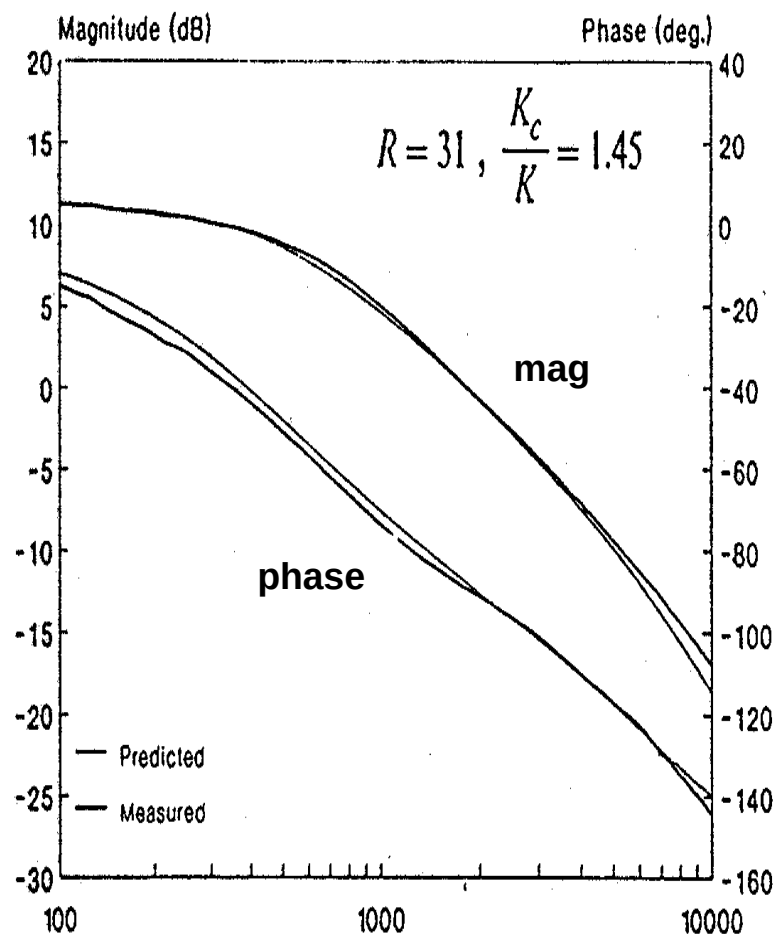
The second pole starts at F_s/π Hz when the converter is operating at the boundary between DCM and CCM and begins to move outwards as the converter enters deeper into DCM.

Experimental Results

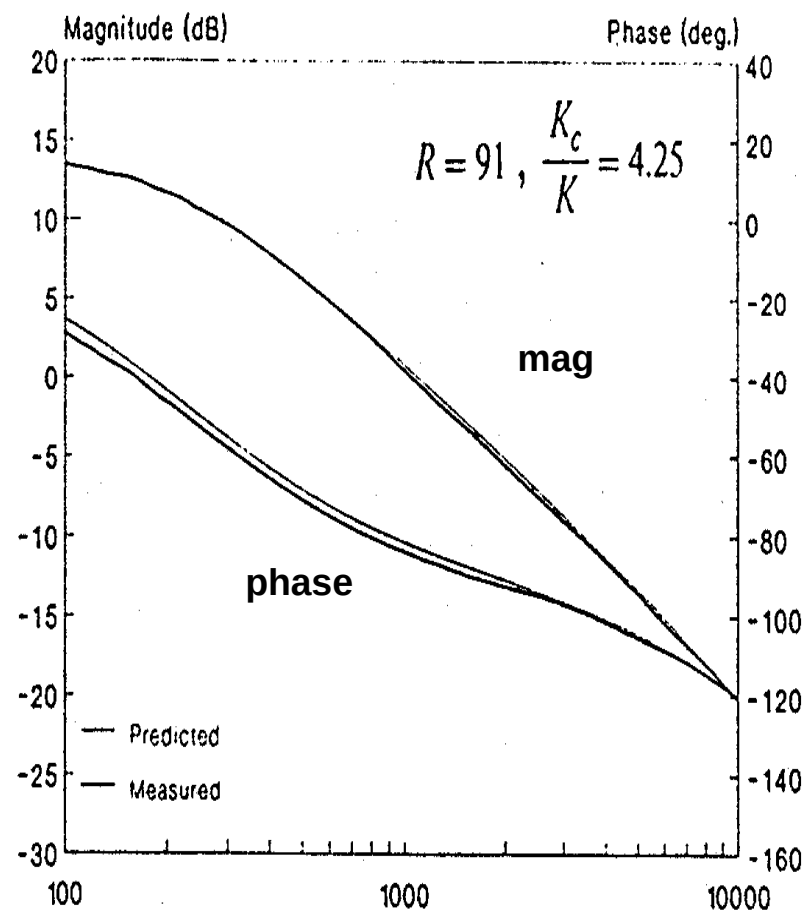
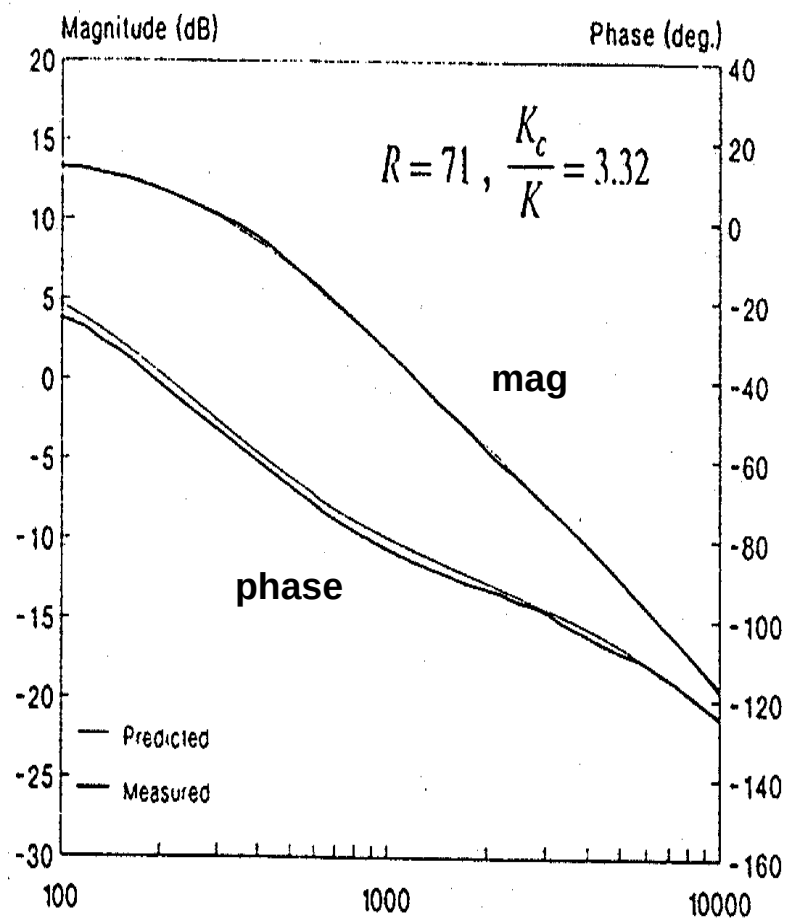
A buck converter was built with the following circuit parameters:

$$\begin{aligned} L_f &= 380\mu H & C_f &= 29\mu F \\ r_{Lf} &= .12\Omega & D &= 0.24 \\ F_s &= 20\text{kHz} & R &= 31, 71, 91\Omega \end{aligned}$$

The phase response shows that the converter in DCM is still a second-order system and not a first order system.



Experimental Results



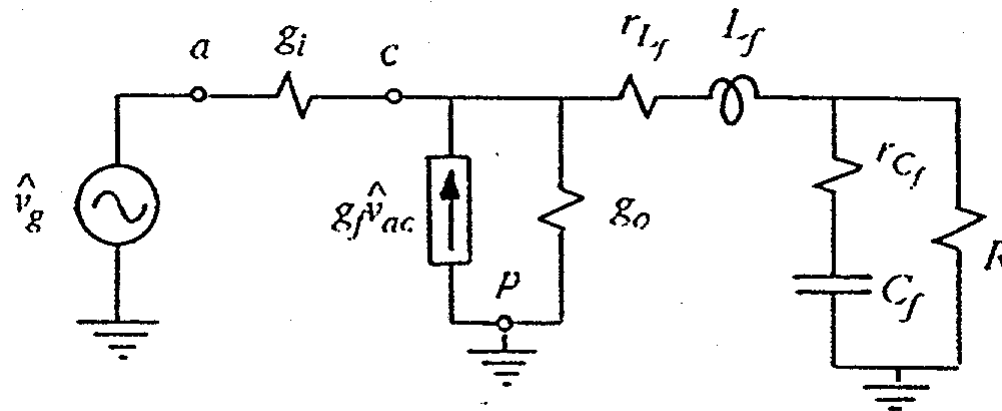
Open-Loop Line-to-Output Transfer Function

To perform this analysis, we replace V_g with a small-signal perturbation v_g and **remove the duty-ratio controlled sources $k_i d$ and $k_o d$ from the model** of PWM switch as shown below.

The line-to-output transfer function has exactly the same form as the control-to-output transfer function in Eq.(21) except that the low-frequency asymptote is given by M :

$$\frac{\hat{V}_o}{\hat{V}_g} = M \frac{1 + s/s_{z1}}{1 + a_1 s + a_2 s^2} \quad (28)$$

where M , s_{z1} , a_1 and a_2 are given in Eqs.(16) and (22b,c,d).



The Boost Converter in DCM

The control-to-output transfer function of the boost converter in dcm can be shown to be given by:

$$\frac{\hat{v}_o}{\hat{d}} = H_d \frac{(1 + s/s_{z1})(1 - s/s_{z2})}{(1 + s/s_{p1})(1 + s/s_{p2})} \quad (47)$$

Where s_{z1} is the usual zero due to the output filter capacitor given in Eq.(36) and

$$s_{z2} = \frac{R}{M^2 L_f} \quad (48a)$$

$$s_{p1} = \frac{2M-1}{M-1} \frac{1}{RC_f} \quad (48b)$$

$$s_{p2} = 2F_s \left(\frac{1-1/M}{D} \right)^2 \geq 2F_s \quad (48c)$$

$$H_d = \frac{2V_o}{D} \frac{M-1}{2M-1} \quad (48d)$$

We see that the r.h.p. zero has exactly the same dependence on the circuit parameters in DCM as in CCM because the r.h.p. zero in CCM is given by $s_{z2} = RD'^2/L_f$. The relative position of the r.h.p. zero to the second pole can be easily determined and is given by:

$$s_{z2} = \frac{s_{p2}}{1-1/M} > 2F_s \quad (49)$$

Hence, s_{z2} occurs always after the second pole since $M > 1$ for the Boost converter.

The Buck-Boost Converter in DCM

For the Buck-Boost converter the zeroes and the poles of the control-to-output transfer function are given by

$$s_{z2} = \frac{R}{M(1+M)L_f} \quad (50a)$$

$$s_{p1} = \frac{2}{RC_f} \quad (50b)$$

$$s_{p2} = 2F_s \left(\frac{1/D}{1+1/M} \right)^2 \geq 2F_s \quad (50c)$$

$$H_d = \frac{V_g}{\sqrt{K}} \quad (50d)$$

Again it can be easily seen that the r.h.p. zero has the same dependence on M and the circuit parameters as the r.h.p. zero in CCM. The relative position of s_{z2} to s_{p2} is given by

$$s_{z2} = s_{p2} \left(1 + 1/M \right) > 2F_s \quad (51)$$

Hence, as in the case of the Boost converter, s_{z2} occurs after s_{p2} .

Design Compensator, based on CCM as it is still safe if the system works to DCM mode (phase margin on double pole is dangerous)

References

[1] Vatche Vorperian, “Simplified analysis of PWM converters using model of PWM Switch Part 1: Continuous Conduction Mode”, IEEE Transactions on Aerospace and Electronic System, Vol.26, No.3 May 1990, pp. 490-496.

[2] Vatche Vorperian, “Simplified analysis of PWM converters using model of PWM Switch Part 2: Discontinuous Conduction Mode”, IEEE Transactions on Aerospace and Electronic System, Vol.26, No.3 May 1990, pp. 497-505.